

Early Prediction of Pediatric Sepsis Using Machine Learning, Deep Learning, and Explainable AI on the PIC Database: A Comparative Study Using Phoenix Sepsis Criteria

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Abstract

Sepsis remains a leading cause of paediatric intensive care mortality, and existing prediction models are largely trained on adult populations using outdated diagnostic definitions, leaving a gap for paediatric-specific, reproducible work on open data. This study develops a hybrid framework for early prediction of paediatric sepsis on the Pediatric Intensive Care (PIC) database, using the Phoenix Sepsis Criteria as the labelling standard. Using PIC database version 1.1.0 ($n = 3,163$ infection-restricted patients), a systematic leakage audit across 306 candidate features removed vasoactive medication indicators and measurement-frequency features carrying label information by construction. The resulting 225-feature dataset was trained using six baseline classifiers, a tuned CatBoost model, three deep learning architectures (FT-Transformer, ResNet, TabNet), a CatBoost–FT-Transformer stacking ensemble (the CBFTTS-SA model), and a CTGAN-based synthetic augmentation pipeline with Iman–Conover correlation correction. Tuned CatBoost achieved AUROC 0.9868 (AUPRC 0.9691, Brier Score 0.0385) at a recommended threshold of 0.365, corresponding to 90% sensitivity and 96.3% specificity. The CBFTTS-SA model achieved a statistically indistinguishable AUROC of 0.9851 (DeLong $p = 0.388$) and is presented as the study's forward-looking architecture, since transformer-based models are expected to close the gap with gradient boosting as training data scales; CatBoost remains the stronger, fully interpretable choice at the present cohort size. DeLong tests confirmed CatBoost's significant superiority over five non-CatBoost baselines; FT-Transformer alone was statistically non-inferior ($p = 0.169$) but with substantially lower specificity (0.852 vs 0.963). TreeSHAP analysis identified lactate_max, D-dimer_max, and platelets_min as dominant predictors, mapping onto Phoenix cardiovascular and coagulation domains. Subgroup analysis confirmed robust generalisation across all paediatric age bands (AUROC 0.970–1.000). To the best of our knowledge, this is the first published study combining CatBoost, Phoenix labelling, SHAP explainability, and a CatBoost–Transformer stacking ensemble on the PIC database in one reproducible pipeline.

Keywords: pediatric sepsis, machine learning, CatBoost, Phoenix Sepsis Criteria, SHAP, explainable AI, PIC database, stacking ensemble, synthetic data augmentation

1. Introduction

Sepsis is a life-threatening condition in which the immune system's response to infection damages the body's own tissues and organs rather than resolving the infection; in children it can progress from suspected infection to multi-organ failure within hours, and delays in recognition of even an hour have been associated with measurable increases in mortality a relationship first quantified in adult septic shock [1] and corroborated in paediatric critical care analyses [2]. Globally, sepsis is one of the most fatal conditions in paediatric intensive care units (PICUs), with an estimated 25 million paediatric sepsis cases annually and in-hospital mortality ranging from roughly 7% for sepsis to over 33% for septic shock in lower-resource settings [3, 2]. This scale of burden, combined with the speed at which

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children deteriorate, creates a genuine clinical need for prediction systems that can flag at-risk patients before organ failure becomes irreversible.

Children are not small adults with respect to sepsis. Their immune systems, organs, and vital-sign norms are still maturing, and thresholds that indicate danger in adults may fall within the normal range for young children [3]. Blood pressure, respiratory rate, and neurological baselines all differ by age. The bulk of published sepsis ML models do not account for this: they are trained on adult ICU data, most commonly MIMIC-III and MIMIC-IV, and these representational assumptions do not transfer cleanly to children [4, 5]. Pediatric-specific work on open datasets, with rigorous leakage controls and a clinically current labelling standard, remains rare.

For years, the diagnostic standard was SIRS (Systemic Inflammatory Response Syndrome) criteria, adapted for paediatric populations by Goldstein et al. [6], which labels sepsis when a child shows abnormality in two or more of heart rate, respiratory rate, temperature, and white blood cell count. SIRS is non-specific: fever from a cold, post-surgical inflammation, and crying can all trigger its thresholds without any underlying infection, producing a high false-positive rate that leads to alert fatigue and unnecessary interventions. PELOD-2 (Pediatric Logistic Organ Dysfunction score) improved on this for measuring organ-dysfunction severity in critically ill children [7], but it was designed to assess existing dysfunction rather than predict sepsis onset. The Phoenix Sepsis Score, developed by Sanchez-Pinto et al. and validated through an international consensus effort by Schlapbach et al., replaces the proxy-based SIRS approach with a direct measure of organ dysfunction across four domains respiratory, cardiovascular, coagulation, and neurological validated against over three million patient encounters [3, 2]. A total Phoenix score of two or above indicates sepsis. This study uses Phoenix as its labelling standard because it is the only definition that operationalises actual organ failure rather than systemic inflammation as a stand-in [3].

An important methodological constraint must be acknowledged up front. The Pediatric Intensive Care (PIC) database, the data source for this study, does not record the fraction of inspired oxygen (FiO_2) or mechanical ventilation parameters, making the respiratory Phoenix component non-computable; the absence of Glasgow Coma Scale (GCS) documentation similarly renders the neurological component zero throughout the cohort [8]. Sepsis classification in this study therefore rests on the cardiovascular and coagulation Phoenix domains alone, a limitation shared with other machine learning studies applying Phoenix criteria to datasets with incomplete physiological coverage [9]. This does not undermine the validity of the labelling approach, given that cardiovascular instability and coagulation dysfunction are consistently identified as the dominant organ-failure signatures driving mortality and deterioration in paediatric sepsis.

Machine learning applied to sepsis prediction has expanded considerably over the past decade: Fleuren et al. catalogued 130 published models across 28 studies in a single systematic review, and Yang et al. examined 23 eligible studies with a narrower methodological lens, both arriving at the conclusion that predictive performance has improved with richer feature sets and more sophisticated algorithms [10, 5]. This body of work carries two structural problems. The first is dataset bias: MIMIC-III and MIMIC-IV dominate training across the field, and adult ICU physiology does not transfer cleanly to children, whose age-dependent vital-sign ranges, immune responses, and organ dysfunction signatures differ substantially from adult norms [5, 4]. The second is methodological weakness: Deng et al. showed that data leakage, inconsistent outcome definitions, and weak validation designs appear persistently across clinical ML studies, overstating the actual predictive capability of many published models [11]. Le et al. trained one of the earlier paediatric-specific models on UCSF PICU data, reporting an AUROC of 0.916 at the time of onset with prediction up to four hours before clinical sepsis onset, establishing that early prediction was achievable in children, though replication across hospital systems has been limited [4].

Among architectures applied to structured clinical data, gradient boosted decision trees have repeatedly come out on top. CatBoost, XGBoost, and LightGBM handle non-linear feature interactions and mixed variable types without the preprocessing overhead neural networks require, and their native class-imbalance handling suits ICU datasets where positive cases are rare, outperforming deep learning on moderate-sized tabular datasets repeatedly [12, 9]. This preference has been confirmed empirically in the paediatric sepsis literature. Chanci et al. applied CatBoost with Phoenix Sepsis Criteria labelling on a US hospital dataset and reported an AUROC of 0.98, including a SHAP explainability analysis, though the underlying data remains proprietary and inaccessible for external validation [9]. Han et al. compared five tree-based algorithms – extra trees, random forest, LightGBM, XGBoost, and CatBoost – for paediatric bloodstream infection prediction, with random forest achieving the best performance (AUROC 0.874 on the derivation set, 0.762 on external validation) [13]. Shi et al. applied an XGBoost and GRU-based recurrent neural network framework with SHAP interpretability to predict sepsis in paediatric patients transitioning from the

emergency department to the PICU, achieving an AUROC of 0.915 [14].

Two problems recur consistently enough across the sepsis ML literature to warrant treatment as active design decisions rather than background noise. The first is class imbalance: in the cohort used here, sepsis-positive patients are outnumbered roughly three to one. SMOTE and other synthetic oversampling strategies are the standard response, but t-SNE visualisation of this dataset’s feature space showed substantial class overlap between sepsis-positive and sepsis-negative patients, and generating synthetic minority samples in a region where the classes are not cleanly separable risks inflating performance metrics without improving genuine discriminative capability; class weights were used instead, a standard approach for clinical ML [11]. The second is leakage, which Deng et al. identified as widespread and underreported across clinical ML studies [11]; a systematic pre-training audit of candidate features was conducted in this study, detailed in Section 2.1.4.

Deep learning has been applied to sepsis prediction with meaningful success, though much of that success has come from sequential or time-series formulations. Lauritsen et al. trained LSTM-based networks on electronic health record sequences [15]; Moor et al. applied Gaussian Process Temporal Convolutional Networks with dynamic time warping to early sepsis recognition [16]; Kamaleswaran et al. used convolutional neural networks to predict sepsis up to eight hours before onset in a paediatric cohort [17]. These results depend on longitudinal data structures that are not always available or consistently recorded across institutions.

For purely tabular data, two architectures have attracted more recent attention. Gorishniy et al. introduced the Feature Tokenizer Transformer (FT-Transformer), which treats each tabular feature as an individual token processed through multi-head self-attention, outperforming prior deep learning methods across multiple tabular benchmarks while remaining competitive with gradient boosted trees [12]. Arik and Pfister introduced TabNet, which uses sequential attention to select features instance by instance, giving it built-in interpretability without post-hoc methods [18]. This study benchmarks both architectures, plus a ResNet baseline, against tuned CatBoost, and combines CatBoost with the FT-Transformer in a stacking ensemble named the **CBFTTS-SA model** in this study that forms the paper’s featured architecture.

The PIC training cohort used here contains 2,530 patients, a sample size that is workable but small enough to raise the question of whether synthetic augmentation could improve model robustness. CTGAN, introduced by Xu et al., is a conditional generative adversarial network for tabular data using mode-specific normalisation and conditional sampling [19], but it is well documented to generate features from marginal distributions independently, destroying correlation structure [19]. In this dataset the problem was severe: the Spearman correlation between INR and PT in real patients was 0.999, reflecting their shared biological pathway, while vanilla CTGAN reduced this to 0.016. A three-stage correction pipeline was developed for this study: CTGAN trained separately per class, Iman-Conover rank correlation injection [20] to restore the real correlation structure, and a zero-inflation correction for heavy-zero-mass features.

A model that performs well on a held-out test set is not automatically useful in a clinical environment; trust requires that the person acting on a prediction can understand why it was made. A false positive sepsis alert carries real costs (unnecessary antibiotics, invasive procedures, alert fatigue), and a missed true positive can be fatal, so asking a clinician to act on an unexplained probability score is asking them to outsource judgement to a system they cannot interrogate [21]. Regulatory frameworks such as the EU AI Act now classify high-risk medical AI in ways that effectively require interpretability as a condition of deployment. Among post-hoc explanation methods, SHAP has become the most widely adopted: Lundberg and Lee grounded feature attribution in Shapley values from cooperative game theory [22], and SHAP is the only additive attribution method that simultaneously satisfies the local accuracy, missingness, and consistency axioms [22]. For tree-based models, the TreeSHAP algorithm computes exact Shapley values without sampling approximation. Prior work on the PIC database has already demonstrated that SHAP surfaces clinically meaningful patterns from Chinese PICU data: Shen et al. applied CatBoost with SHAP for AKI mortality prediction [23], Zhou et al. used SHAP for DIC prediction [24], and Huang et al. incorporated SHAP into a sepsis survival analysis [25]. Outside the PIC context, Yang et al. applied explainable ensemble ML with SHAP to MIMIC-IV sepsis data to identify clinically relevant feature contributions [26].

The PIC database, described by Zeng et al., is the first publicly accessible intensive care database drawn from a Chinese paediatric population, maintained by the Children’s Hospital of Zhejiang University School of Medicine [8]. Its public availability makes it one of the few resources outside North America and Europe on which paediatric clinical ML models can be openly developed and validated, and it has been used across a range of prediction tasks: sepsis survival analysis [25], AKI mortality with CatBoost and SHAP [23], DIC prediction [24], ensemble-based ICU

mortality prediction [27], and the triglyceride-glucose index in paediatric sepsis mortality [28]. Cohort construction and database characteristics for the present study are detailed in Section 2.1.

Despite this volume of research, a specific and reproducible combination has not appeared in the published literature. Three points illustrate the gap most directly:

- Chanci et al. applied CatBoost with Phoenix Sepsis Criteria labelling and reached strong predictive performance (AUROC 0.98) with a SHAP explainability layer, but the work relied on proprietary US hospital data, so neither the dataset nor the trained model is accessible for external validation [9].
- Shen et al. used CatBoost with SHAP on the PIC database, but the target outcome was acute kidney injury mortality rather than sepsis onset, so the clinical domain and labelling logic are entirely different [23].
- Huang et al. worked with PIC data on a sepsis-related question but framed it as survival analysis, without Phoenix criteria for case definition and without any explainability component [25].

No published study has brought all four elements together in a single pipeline: CatBoost as a primary model, Phoenix Sepsis Criteria as the labelling standard, SHAP-based explainability, and the PIC database as the data source. Each missing element in the prior work represents a real limitation, whether in reproducibility, clinical validity of the outcome definition, or the absence of an interpretive layer clinicians can engage with.

This study fills that gap with a complete machine learning, deep learning, and explainable AI pipeline for early paediatric sepsis prediction on the PIC database, using Phoenix Sepsis Criteria as the labelling standard throughout. To the best of the authors’ knowledge, this is the first published study combining CatBoost, Phoenix Sepsis Criteria labelling, SHAP explainability, and the PIC database in a single pipeline; as authors of this work, we developed a hybrid framework composed of the following four components:

- A systematic leakage audit in which 306 candidate features were audited before any model was trained, identifying and removing features that carried label information by construction.
- A principled deep learning benchmark in which three architectures FT-Transformer, ResNet, and TabNet were benchmarked against CatBoost using DeLong significance testing, and CatBoost was combined with the FT-Transformer in a stacking ensemble, named the **CBFTTS-SA model** in this study and presented as the paper’s featured architecture, on the basis that transformer-class models are expected to close the gap with gradient boosting as training data scales beyond this cohort’s moderate size.
- A synthetic data augmentation study in which CTGAN [19] was extended with Iman-Conover rank correlation injection [20] to address CTGAN’s known correlation-destruction problem, and evaluated honestly against real-data training.
- A SHAP interaction analysis in which a lactate–platelets axis and an INR–D-dimer coagulopathy axis were identified, mapping directly onto the cardiovascular and coagulation Phoenix domains and providing biological grounding for the CatBoost component’s predictions.
- A performance comparison in which tuned CatBoost is the strongest and fully interpretable model at the present cohort size (AUROC 0.9868), while the CBFTTS-SA model is statistically indistinguishable from it (AUROC 0.9851, $p = 0.388$) and is identified as the architecture with the greatest scaling potential, on the basis that transformer-class models are expected to close the gap with gradient boosting as training data grows beyond this cohort’s moderate size leading to the conclusion that CatBoost remains the stronger choice for immediate deployment today, but the CBFTTS-SA model is positioned to become the superior architecture as larger paediatric ICU datasets become available.

The remainder of this paper is organised as follows: Section 2 describes the dataset, preprocessing, and methods; Section 3 presents experimental results; Section 4 discusses findings and limitations; and Section 5 concludes.

2. Materials and Methods

2.1. Data Pipeline Overview

Figure 1 gives a visual overview of the six-stage cohort-construction and preprocessing pipeline described in this subsection, from the raw PIC tables through to the model-ready dataset. The stepwise counts shown in the figure are under final reconciliation against Table 2 and the text below; the entry counts, intermediate cohort sizes, and sepsis-prevalence figures reported in prose throughout this subsection are the authoritative values for this study.

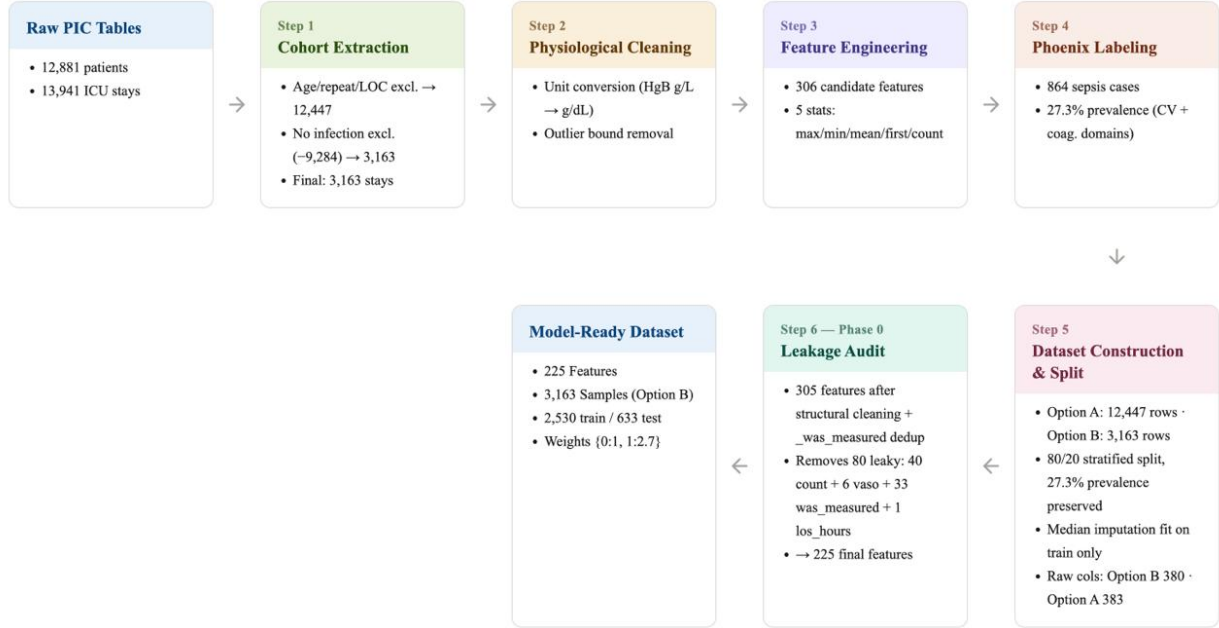


Figure 1: Overview of the cohort-construction and preprocessing pipeline, from raw PIC tables through cohort extraction, physiological cleaning, feature engineering, Phoenix labelling, dataset construction, and the train/test split with median imputation, yielding the final model-ready dataset.

2.1.1. Data Source

This study uses the Pediatric Intensive Care (PIC) database (version 1.1.0), a publicly available, de-identified collection of electronic health records from paediatric patients admitted to the intensive care unit at the Children’s Hospital of Zhejiang University School of Medicine in China [8]. The dataset is accessible to credentialed researchers through PhysioNet and spans routine clinical care collected between 2010 and 2018, covering patient demographics, laboratory measurements, vital signs, medications, microbiology results, fluid balance, and clinical symptoms. All timestamps are shifted to preserve patient privacy while maintaining the relative temporal ordering of events, making valid time-series analysis possible.

The database contains 12,881 unique patients across 13,941 ICU stays and is one of the largest publicly available paediatric intensive care datasets. Its relational structure mirrors that of the widely benchmarked MIMIC database, adapted for a paediatric population. Table 1 summarises the key characteristics of the PIC database.

To construct the feature set required for early sepsis onset prediction, data were extracted from multiple relational tables within the PIC database and linked using patient and ICU stay identifiers to form a unified representation of each admission. The primary clinical tables used were: PATIENTS, ADMISSIONS, and ICUSTAYS for demographics and admission metadata; LABEVENTS and CHARTEVENTS for laboratory results and physiological measurements; PRESCRIPTIONS, INPUTEVENTS, and OUTPUTEVENTS for treatment and fluid records; MICROBIOLOGYEVENTS for infection-related data; and EMR_SYMPTOMS and SURGERY_INFO for clinical context. Three reference tables D_ITEMS, D_LABITEMS, and D_ICD_DIAGNOSES were used for variable mapping only and did not contribute directly to the feature matrices.

Table 1: Summary characteristics of the PIC database v1.1.0.

Attribute	Value
Dataset	Pediatric Intensive Care (PIC)
Version	v1.1.0
Total patients	12,881
Total ICU stays	13,941
Time period	2010–2018
Population	Paediatric (age 0–18)
Data type	EHR (labs, vitals, meds, symptoms)
Clinical tables	11 (+ 3 reference tables)
Source	Children’s Hospital, Zhejiang University
Access	PhysioNet (credentialed)

2.1.2. Cohort Definition

The initial pool comprised 13,941 ICU stays from 12,881 unique paediatric patients. A sequential filtering and restriction process was applied to define the analysis cohort. Stays where the patient’s age fell outside the range of 0 to 18 years were excluded ($n = 42$). Repeat ICU admissions were removed, retaining only each patient’s first stay to ensure observation independence ($n = 1,060$ excluded). Stays with a total ICU length of stay under six hours were excluded as they provide insufficient physiological observation time for meaningful feature extraction ($n = 392$). The resulting intermediate cohort comprised 12,447 ICU stays. A further restriction to patients with clinically suspected infection was then applied, as the prediction model is intended for deployment after infection suspicion rather than across all PICU admissions indiscriminately ($n = 9,284$ excluded). The final modelling cohort comprised 3,163 stays. Table 2 details each step. The majority of the initial exclusions arose from repeat ICU admissions (7.6%), removed to preserve statistical independence of observations; the restriction to patients with suspected infection (74.6% of the intermediate cohort) reflects the intended clinical deployment context of the model.

Table 2: Cohort construction: stepwise exclusions and restriction from the PIC database.

Step	Count	% of Total
Initial ICU stays	13,941	100.0%
Excluded: age out of range	42	0.3%
Excluded: non-first ICU stay	1,060	7.6%
Excluded: LOS < 6 hours	392	2.8%
Intermediate cohort	12,447	89.3%
Excluded: no suspected infection	9,284	66.6%
Final modelling cohort	3,163	22.7%

2.1.3. Preprocessing Pipeline

Data preprocessing was implemented as a six-stage modular pipeline cohort filtering, physiological data cleaning and validation, feature extraction, sepsis label definition, label verification and dataset construction, and train-test split with imputation with each stage producing intermediate output and logs to enable full reproducibility. This stage corresponds to the preprocessing block of the overall study pipeline shown in Figure 3.

Physiological data cleaning and validation. The objective of this stage was to remove implausible values arising from measurement or transcription errors while preserving genuine extreme clinical observations in critically ill patients. For each variable, physiologically informed upper and lower bounds were defined with conservative buffers based on reported clinical extremes in the paediatric critical care literature [3, 8]; values outside these bounds were treated as missing rather than corrected, avoiding distortion of subsequent statistical summaries. Three systematic data issues were identified and corrected iteratively: haemoglobin values stored in g/L were being compared against

g/dL thresholds, invalidating 87.9% of readings, and a unit conversion factor was applied to resolve this; the anion gap lower bound was revised from 0 to -5 mmol/L to accommodate measurement and calculation artefacts; and the SpO₂ upper bound was raised from 100.0% to 100.5% to accommodate analyser rounding artefacts.

Feature extraction. Feature extraction was performed over a fixed 24-hour window following ICU admission, chosen because it aligns with the natural rhythm of clinical assessment cycles in PICUs and is consistent with the prediction horizon used in prior paediatric sepsis ML studies [4, 9]. All predictors are computed from data within this window, ensuring they temporally precede the prediction target and eliminating any possibility of future-data leakage. For each physiological variable, five statistical summaries were computed (maximum, minimum, mean, first recorded value, measurement count); beyond these, domain-informed engineering produced trend features (difference between first and last recorded values), derived physiological metrics (shock index, mean arterial pressure), age-adjusted vital-sign abnormality indicators, fluid balance features, infection-related features derived from antibiotic and microbiology records, 31 binary symptom indicators, and surgical context indicators. This stage produced 306 candidate feature columns. Structural cleaning – removal of identifier and free-text columns, zero-variance fields, redundant unit-duplicate columns, Phoenix component score columns, and deduplication of `_was_measured` indicators to one column per variable – reduced this to a 305-feature dataset prior to the leakage audit described below. The leakage audit (Section 2.1.4) removed 80 of these features in a single pass, yielding the final 225-feature clean modelling set used throughout this study.

2.1.4. Leakage Audit

Prior to model training, a systematic leakage audit was conducted across the 305-feature structurally-cleaned dataset (Section 2.1.3). Six candidate leakage suspects were examined. Four were assessed by training an isolated classifier on each candidate feature group alone and measuring its stand-alone discriminative power (isolation AUROC): a group whose isolation AUROC approaches the full-model AUROC despite representing only a narrow, clinically indirect signal is evidence that the group encodes label information by construction rather than genuine physiology.

Measurement frequency features (`_count` columns, $n = 40$) were found to be directly correlated with the Phoenix labelling process: sepsis-positive patients had systematically higher measurement counts, driven by clinical urgency rather than physiology (isolation AUROC 0.878). Vasoactive medication indicators ($n = 6$ columns) were the most structurally leaky category: 73.5% of sepsis patients received vasoactives compared to 23.7% of non-sepsis patients, and these variables are themselves components of the Phoenix cardiovascular domain that defines the label (isolation AUROC 0.863). Trend features ($n = 16$) and fluid balance variables ($n = 4$) were assessed and retained: trend features showed a maximum feature-label correlation of 0.159 and isolation AUROC of 0.848, indicative of genuine physiological signal, while fluid balance features achieved an isolation AUROC of only 0.609, confirming no leakage concern.

Two further diagnostic checks were performed. The deduplicated `_was_measured` missingness indicators ($n = 33$, one retained per variable after the structural-cleaning deduplication) showed the same physician-ordering signature as the count columns (e.g. $r = 0.38$ for `inr_trend_was_measured`, `fibrinogen_max_was_measured`, and `ddimer_max_was_measured` against the sepsis label) and were removed in full alongside them. A comparison of training-set imputation medians against full-cohort medians, run as a diagnostic for outcome-dependent features, flagged `los_hours` (ICU length of stay) with the largest discrepancy (4.165 hours); this column was identified as an outcome variable unknowable at the time of prediction and was dropped. A companion patient-overlap check between training and test partitions could not be directly verified because patient identifiers were removed during preprocessing; overlap is instead structurally prevented by the cohort-definition rule restricting the cohort to each patient’s first ICU stay (Section 2.1.2).

In total, 80 features across four leaky categories (`_count`, vasoactive medication, `_was_measured`, and `los_hours`) were removed, yielding the final clean feature set of 225 features. A post-cleaning isolation test on this reduced set achieved AUROC 0.978 and AUPRC 0.950, confirming that the retained features carry strong legitimate predictive signal.

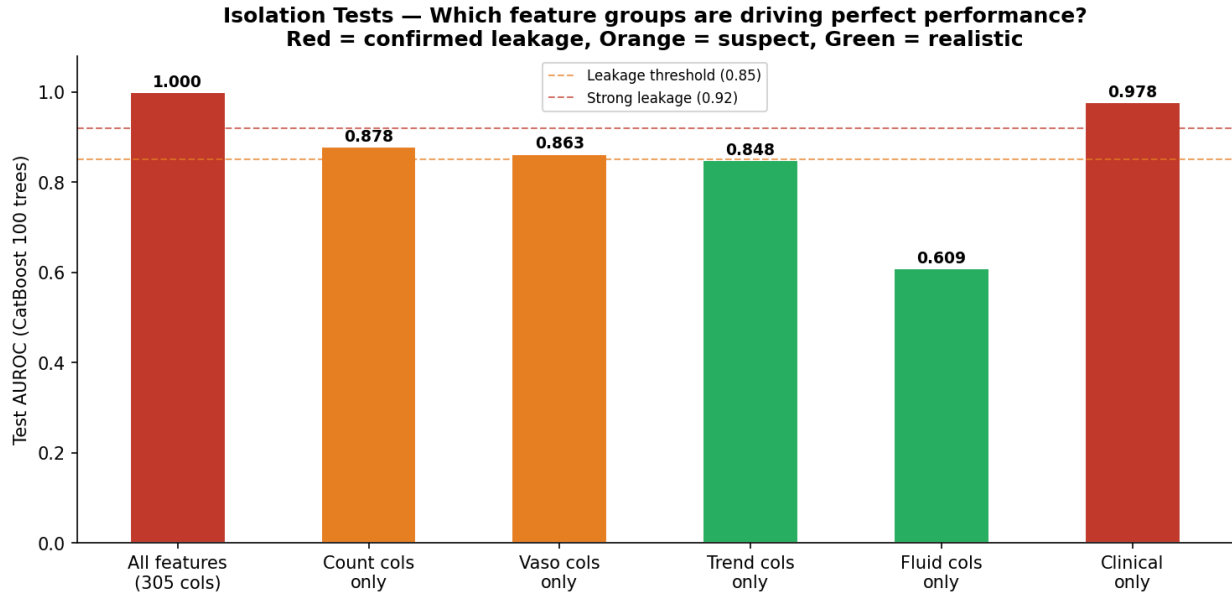


Figure 2: Isolation AUROC values of each leakage group.

2.1.5. Sepsis Label Definition (Phoenix Criteria)

Sepsis labels were defined using the Phoenix paediatric sepsis criteria [3]. As established in Section 1, the respiratory and neurological Phoenix components were non-computable in the PIC database (FiO₂, ventilation parameters, and GCS are absent) and were set to zero; labelling therefore rested on the cardiovascular and coagulation Phoenix domains exclusively, consistent with other PIC-based studies facing the same data constraint [9]. Missing values within the computable domains were treated as contributing zero points, consistent with the Phoenix framework’s convention that absent measurements contribute no positive score [3]. The final sepsis label was assigned to any patient meeting both conditions: clinically suspected infection and a Phoenix score of two or above.

Applying this definition to the infection-restricted cohort of 3,163 stays identified 864 sepsis cases (27.3%) and 2,299 patients with suspected infection but no sepsis (72.7%). For context, across the full 12,447-stay intermediate cohort, 864 sepsis cases represent 6.9% prevalence, with 2,299 suspected-infection/no-sepsis patients (18.5%) and 9,284 patients with no suspected infection (74.6%).

2.1.6. Dataset Construction and Train-Test Split

A dedicated validation stage confirmed the clinical plausibility of the derived labels. Comparative analysis between sepsis-positive and sepsis-negative patients showed expected clinical patterns: higher lactate levels, lower platelet counts, greater vasoactive medication use, and substantially higher in-hospital mortality in the sepsis group (18.1% versus 4.4%), consistent with the pathophysiology of sepsis and providing face validity for the labelling approach.

Two datasets were constructed from the labelled cohort. Option A encompassed the full 12,447-stay intermediate cohort ($\{0 : 1.0, 1 : 13.4\}$ class weights, reflecting 6.9% prevalence). Option B restricted the population to patients with suspected infection, yielding 3,163 stays ($\{0 : 1.0, 1 : 2.7\}$ class weights, reflecting 27.3% prevalence). **Option B was used for all primary model training, tuning, and evaluation throughout this study; Option A is used exclusively for the cohort-definition sensitivity analysis reported in Section 3.9, in which the primary model architecture is retrained from scratch on Option A to test whether the study’s conclusions are an artefact of the infection-restriction design choice.** Option B was selected as the primary modelling dataset because it reflects the realistic clinical deployment context: the prediction model would be applied after infection has been suspected, not indiscriminately across all PICU admissions.

The Option B dataset was divided into training (80%, $n = 2,530$) and testing (20%, $n = 633$) sets using stratified

sampling to preserve the 27.3% sepsis prevalence across both partitions. Missing values were imputed using median values computed exclusively on the training set and applied to both partitions, preventing any leakage of test-set distributional information into the imputation process. For variables with a missingness rate above 20%, binary indicator features (`_was_measured`) were engineered to capture the clinical signal associated with whether a test was ordered at all, a threshold consistent with common practice in clinical ML studies [11]; after deduplication to one indicator per variable, these were subsequently identified as leaking the same physician-ordering signal as the `_count` features (Section 2.1.4) and were removed from the model inputs in full – no `_was_measured` columns are present in the final 225-feature set. Phoenix score component columns and residual identifier columns were likewise excluded during the leakage-audit stage. The leakage audit (Section 2.1.4) is therefore the sole source of the reduction from the 305-feature structurally-cleaned dataset to the final 225-feature modelling matrix used throughout this study.

2.1.7. Synthetic Data Augmentation

The Option B training cohort of 2,530 patients is workable but modest by deep-learning standards, which raises the question of whether synthetic data could supplement it. To test this, a CTGAN-based synthetic patient generation pipeline was built and evaluated as part of this study, extended with an Iman-Conover rank correlation correction step to address a known limitation of standard CTGAN on clinically correlated variables (Section 1). The pipeline, its design rationale, and the resulting real-data versus synthetic-augmented model comparisons are reported in full in Section 3.7.

2.2. Methods

2.2.1. Experimental Setup

All experiments were conducted on the Option B infection-only cohort (2,530 training / 633 test samples, 27.3% sepsis prevalence), split with stratification to preserve class balance. Missing values were imputed using training-set medians exclusively, applied identically to both partitions. A `StandardScaler` was applied to numerical features; binary indicators for care unit type and gender were excluded from scaling. Class imbalance was addressed using inverse-prevalence class weights ($\{0 : 1.0, 1 : 2.7\}$ for Option B), rather than SMOTE-style synthesis, for the reasons discussed in Section 1. All model training and evaluation was performed in a conda environment using Python 3.10 with CatBoost, XGBoost, LightGBM, scikit-learn, and Optuna. Five-fold stratified cross-validation was used during training to assess generalisation prior to final evaluation on the held-out test set. This study was conducted and reported in accordance with the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) statement for prediction model development [29].

2.2.2. Baseline Models

Six classifiers were trained on the 225-feature clean dataset: CatBoost, XGBoost, LightGBM, Random Forest, Logistic Regression, and a Multi-Layer Perceptron (MLP). All models used five-fold stratified cross-validation, and a fixed operating threshold targeting 85% sensitivity was applied to enable a fair comparison of the precision-recall balance across architectures. The strongest-performing baseline was carried forward for hyperparameter tuning as the primary single-model architecture.

2.2.3. CatBoost Tuning and Alternative Pipelines

CatBoost was selected for tuning based on its superior baseline performance. One hundred Bayesian optimisation trials were conducted using Optuna, with 5-fold cross-validated AUPRC as the objective; AUPRC was preferred over AUROC because it is more sensitive to classifier performance at the clinically relevant high-recall region and less susceptible to inflation under class imbalance. Two alternative single-pipeline architectures were tested against the resulting tuned CatBoost model. First, a tree-based stacking ensemble was constructed with tuned CatBoost, XGBoost, and LightGBM as base learners and a logistic regression meta-learner: ten clinically motivated interaction features (e.g. $\text{lactate}_{\max} \times \text{platelets}_{\min}$, $\text{D-dimer}_{\max} \times \text{INR}_{\max}$) were engineered and added to the base feature set, out-of-fold probabilities from five-fold cross-validation trained the meta-learner, and all base models were retrained on the full training set before test-set evaluation. Second, a Phoenix-aware asymmetric loss function was developed in which false negatives are penalised proportionally to a patient-level severity proxy computed from normalised z-scores of seven Phoenix component variables ($\text{lactate}_{\max}$, $\text{platelets}_{\min}$ (inverted), $\text{INR}_{\max}$, $\text{D-dimer}_{\max}$, $|\text{base_excess}_{\min}|$, PT_{mean} ,

PTT_{max}): the per-sample loss for a positive-class patient is $\text{loss}_i = -w_i \log(p_i)$ with $w_i = \text{base_weight} + \alpha \cdot \text{severity}_i$, and negative-class samples retain standard cross-entropy. Both α and base_weight were jointly optimised with CatBoost hyperparameters via Optuna over 50 trials.

The optimal hyperparameter configuration identified by 100 Optuna trials is presented in Table 3.

Table 3: Optimal hyperparameters identified by Optuna (100 trials, AUPRC-optimised). Best CV AUPRC: 0.9580.

Hyperparameter	Optimal Value
Iterations	510
Learning Rate	0.0342
Tree Depth	7
L2 Leaf Regularisation	3.754
Bagging Temperature	0.975
Random Strength	0.050
Border Count	233
Positive Class Weight	1.820

2.2.4. Primary Model Evaluation Protocol

The tuned CatBoost model was evaluated using five components. Pairwise DeLong tests assessed whether AUROC differences between tuned CatBoost and each comparator (baseline classifiers, deep learning models, alternative pipelines) were statistically meaningful. Probability calibration was assessed using reliability diagrams and Platt scaling. Decision Curve Analysis (DCA) evaluated clinical net benefit across a range of decision thresholds against *treat-all* and *treat-none* reference strategies. A full threshold sensitivity analysis characterised the sensitivity–specificity trade-off across five operating points spanning 80–95% sensitivity, to support an evidence-based clinical deployment recommendation. Finally, subgroup performance was evaluated separately across four clinically defined paediatric age bands neonates (0–28 days), infants (29 days–1 year), children (1–12 years), and adolescents (12–18 years) aligned with established clinical categories in paediatric critical care.

2.2.5. Deep Learning Benchmark

Three deep learning architectures were benchmarked against tuned CatBoost: the Feature Tokenizer Transformer (FT-Transformer), which embeds each tabular feature as an individual token processed through multi-head self-attention [12]; a tabular ResNet, providing a residual feed-forward baseline that isolates what architectural depth alone contributes without attention; and TabNet, which applies sequential instance-wise attention masks to dynamically select a sparse feature subset per patient [18]. Each was trained on the same 225-feature leakage-clean dataset with identical class weights, five-fold stratified cross-validation during hyperparameter search, and the same 80/20 stratified train-test split. Hyperparameters for all three architectures were optimised using Optuna with five-fold cross-validated AUPRC as the objective, matching the CatBoost tuning protocol. Performance is reported at the threshold targeting the 90% sensitivity operating point for each architecture, with pairwise DeLong tests against tuned CatBoost.

2.2.6. CBFTTS-SA Model: CatBoost–FT-Transformer Stacking Ensemble

This study’s featured architecture, named the **CBFTTS-SA model**, stacks tuned CatBoost and the FT-Transformer as base learners with a logistic regression meta-learner trained on their out-of-fold probability outputs from five-fold stratified cross-validation. The FT-Transformer was selected as the second base learner over ResNet and TabNet because it produced the highest standalone AUROC among the three deep learning architectures and its attention-based representation is architecturally most distinct from CatBoost’s ordered-boosting structure, satisfying the standard diversity condition for effective stacking. Two configurations were evaluated: one trained exclusively on the 2,530-patient real-data training set, and a second trained on the hybrid dataset from the synthetic augmentation pipeline (Section 2.2.7), combining all real patients with 10,000 CTGAN-generated patients. In both configurations the meta-learner received only the two base-learner probability columns as inputs, forcing it to act purely as a probability

combiner; base learners were retrained on the full training set before final test-set evaluation. As a secondary experiment, the FT-Transformer’s CLS-token attention weight vector was also extracted and concatenated with the raw 225-feature vector to train an attention-augmented CatBoost variant (450 input dimensions), testing whether the transformer’s learned inter-feature weighting could improve CatBoost’s split-finding directly rather than through stacking.

2.2.7. Synthetic Data Augmentation Pipeline

CTGAN, a conditional generative adversarial network for tabular data [19], was used to generate synthetic patients, with a three-stage correction pipeline developed to address its known correlation-destruction problem (Section 1): CTGAN was trained separately on sepsis-positive and sepsis-negative patients to preserve correct within-class marginal distributions, Iman-Conover rank correlation injection [20] restored the real Spearman correlation matrix within each class, and a zero-inflation correction was applied to variables with heavy zero mass. The final synthetic dataset comprised 10,000 generated patients split proportionally by the training cohort’s 27.3% class prevalence. Three augmentation configurations were evaluated against the real-data CatBoost baseline: a CatBoost model trained exclusively on synthetic data; a hybrid CatBoost trained on all 2,530 real patients (sample weight 4) combined with the 10,000 synthetic patients (sample weight 1); and the hybrid-data CBFTTS-SA model configuration described above. All comparisons used DeLong tests against the real-data tuned CatBoost model, and cross-validation folds used during Optuna tuning remained unweighted to avoid inflating optimisation signal from synthetic samples.

2.2.8. SHAP Explainability Methodology

SHAP explainability analysis was applied to the CatBoost component using the TreeSHAP algorithm, which computes exact Shapley values without sampling approximation [22]. SHAP values were computed for all 633 test-set patients across the full 225 features. Four analyses were performed: global feature importance (ranked by mean absolute SHAP value), dependence analysis for the top features, individual patient explanation via waterfall plots for three clinically informative cases (a true positive, a false negative, and a false positive), and pairwise SHAP interaction analysis for the top ten features using the TreeSHAP interaction API.

2.2.9. Sensitivity Analysis Design

To test whether the study’s conclusions are an artefact of the Option B infection-restriction design choice, the tuned CatBoost architecture with hyperparameters fixed from the Option B optimisation was retrained from scratch on Option A (the full 12,447-stay intermediate cohort, regardless of infection suspicion), with class weights adjusted for its lower sepsis prevalence ($\{0 : 1.0, 1 : 13.4\}$ versus Option B’s $\{0 : 1.0, 1 : 2.7\}$). This design isolates the effect of cohort selection from model architecture and hyperparameter choices: everything except the training population is held constant between the two configurations. All primary training, tuning, and evaluation elsewhere in this study used Option B exclusively; Option A is used only for this sensitivity check, evaluated on the same held-out test set as the primary model.

Figure 3 summarises the overall architecture of the study, from the source database through to explainability analysis. Data flow proceeds from the PIC database through preprocessing, three parallel modelling tracks, evaluation, and SHAP explainability analysis, with each of these stages described in detail in the subsections that follow.

3. Results

3.1. Baseline Model Comparison

All tree-based ensemble methods attained AUROC above 0.97 on the 225-feature clean dataset; gradient boosting methods substantially outperformed Random Forest, Logistic Regression, and MLP. CatBoost produced the strongest performance at baseline (AUROC 0.9863) with the lowest Brier score (0.0399), as summarised in Table 4, and was carried forward for hyperparameter tuning.

The baseline comparison is visualised in Figure 4, displaying ROC and Precision-Recall curves for all six models. The Precision-Recall curve is informative given the 27.3% class prevalence. Gradient boosting methods maintain high precision across the recall range, while Logistic Regression and MLP show early signs of degradation.

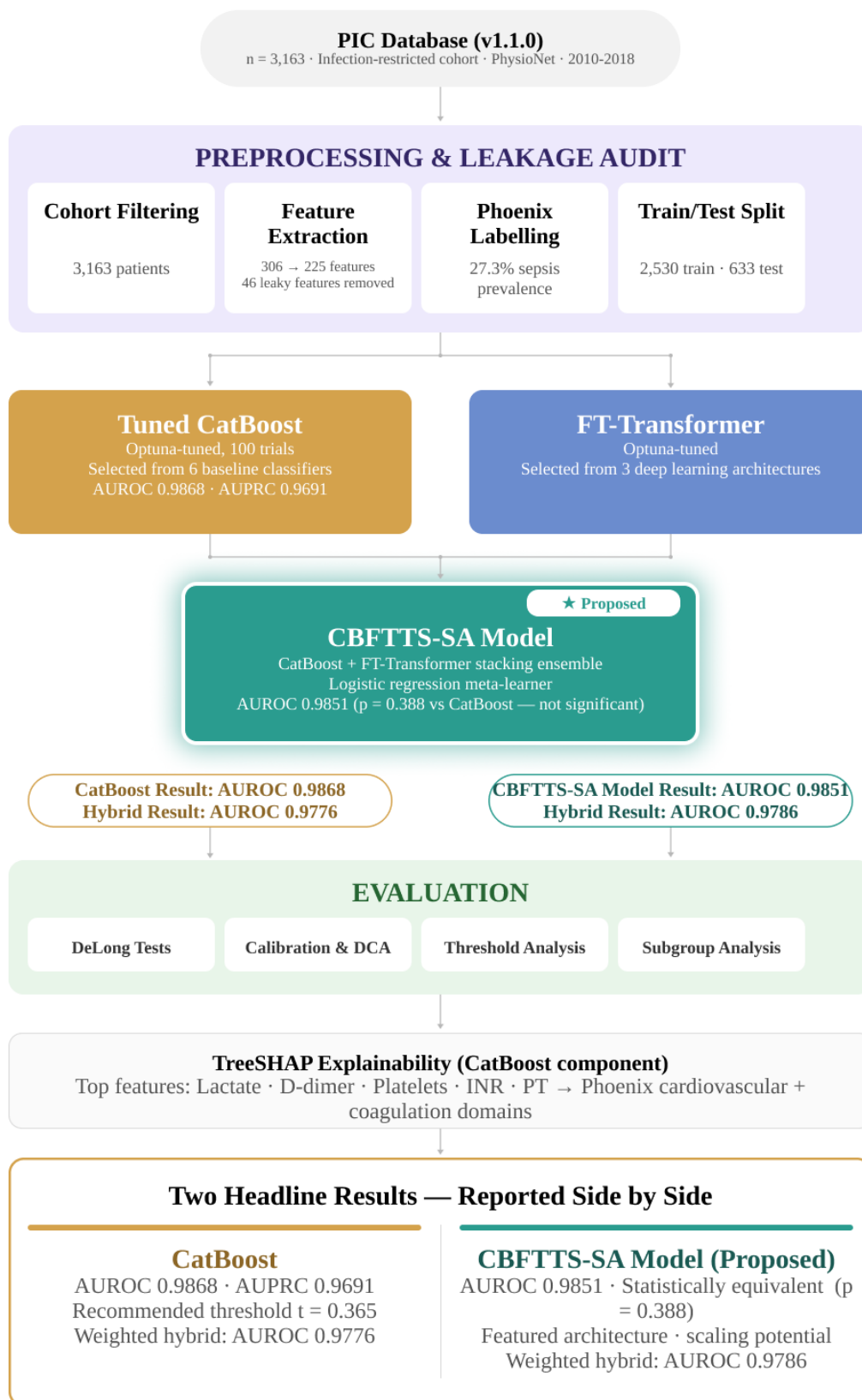


Figure 3: Complete research pipeline for early paediatric sepsis prediction. Data flow proceeds from the PIC database through preprocessing, three parallel modelling tracks, evaluation, and SHAP explainability analysis. The tuned CatBoost model (amber, starred) is the primary single-model architecture selected for downstream evaluation.

Table 4: Baseline model comparison on the Option B held-out test set ($n = 633$), at a decision threshold fixed to 85% sensitivity for cross-architecture comparability. All models were trained with five-fold stratified cross-validation; the metrics reported here are test-set evaluations. Bootstrap 95% CIs not computed for baseline configurations; see Table 12.

Model	AUROC	AUPRC	Sensitivity	Specificity	F1	Brier Score
CatBoost	0.9863	0.9684	0.855	0.980	0.881	0.0399
LightGBM	0.9806	0.9570	0.855	0.963	0.876	0.0521
XGBoost	0.9780	0.9512	0.867	0.963	0.882	0.0505
Random Forest	0.9656	0.9222	0.855	0.928	0.836	0.0712
Logistic Regression	0.9152	0.8477	0.855	0.833	0.744	0.1038
MLP	0.9212	0.8635	0.855	0.863	0.771	0.1045

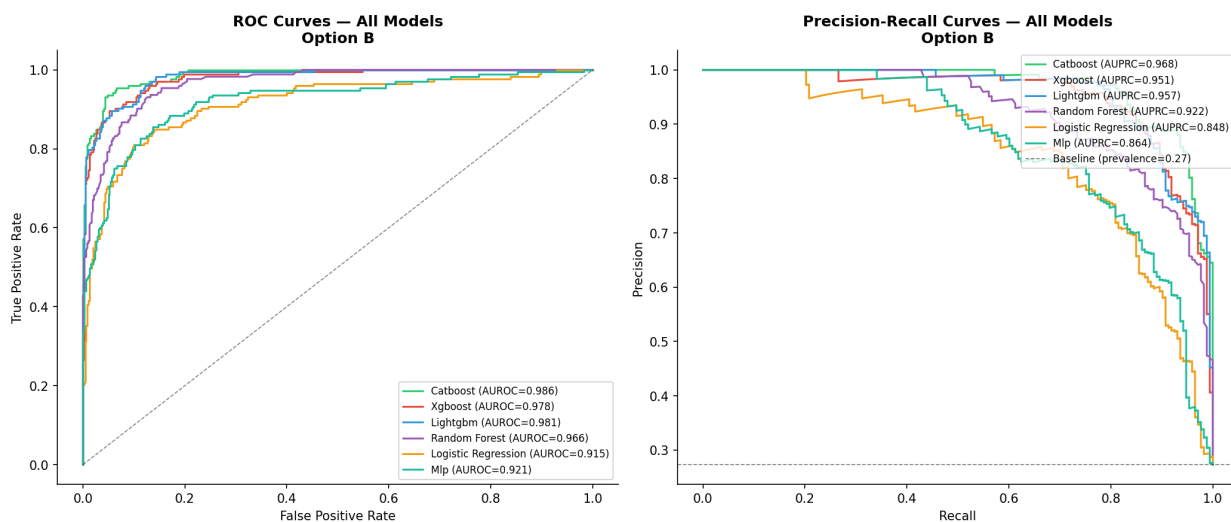


Figure 4: ROC (left) and Precision-Recall (right) curves for all six baseline classifiers on the Option B test set.

3.2. Hyperparameter Tuning CatBoost

The optimal hyperparameter configuration identified by 100 Optuna trials is presented in Table 3 (Section 2.2.3). The tuned model achieved AUROC 0.9868 and AUPRC 0.9691 on the test set. The Optuna optimisation trajectory is shown in Figure 5.

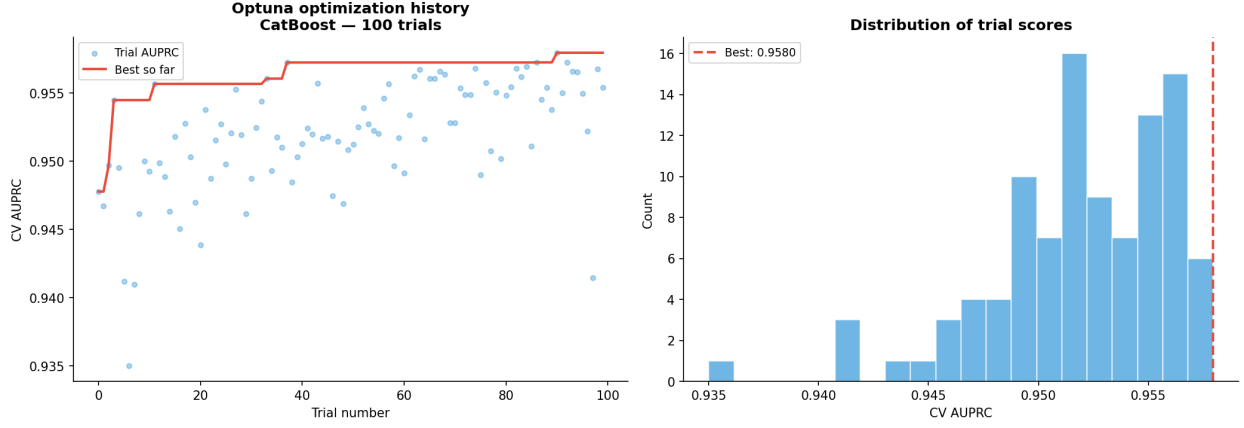


Figure 5: Optuna trial AUPRC values across 100 trials. Convergence is observed by approximately trial 60, with subsequent marginal gains.

3.2.1. Tree-Based Stacking Ensemble (CatBoost + XGBoost + LightGBM)

This tree-based ensemble is a distinct experiment from the CatBoost–FT–Transformer CBFTTS-SA model reported in Section 3.5; it is evaluated here purely as a same-family (all-gradient-boosting) alternative to single-model CatBoost. The ensemble achieved AUROC 0.9851 and AUPRC 0.9680, marginally below tuned CatBoost on both metrics. The DeLong test confirmed the difference was not statistically significant ($p > 0.05$), and the added pipeline complexity yielded no measurable benefit over CatBoost alone.

3.2.2. Phoenix-Aware Custom Loss

Optuna converged to $\alpha = 2.831$ and $\text{base_weight} = 4.467$. Despite its alignment with the clinical objective, the custom loss model underperformed significantly: AUROC 0.9674, AUPRC 0.9313, and Brier Score 0.1005 more than twice that of tuned CatBoost.

3.2.3. Model Selection

Neither alternative pipeline improved on tuned CatBoost, which remains the strongest and most efficient single-model architecture in this study. A three-way comparison is presented in Table 5; the paper’s separately evaluated featured architecture, the CBFTTS-SA model (CatBoost–FT–Transformer stacking ensemble), is reported in Section 3.5.

Table 5: Comparison of tuned CatBoost against the two alternative single-family pipelines on the Option B test set ($n = 633$) at the recommended threshold ($t = 0.365$). Bootstrap 95% CIs were not computed for these configurations; point estimates are reported.

Configuration	AUROC	AUPRC	Sensitivity	Specificity	F1	Brier Score
Tuned CatBoost (primary single model)	0.9868	0.9691	0.896	0.963	0.899	0.0385
Tree-Based Stacking Ensemble	0.9851	0.9680	0.855	0.978	0.894	0.0422
Custom Loss (Phoenix-aware)	0.9674	0.9313	0.855	0.930	0.839	0.1005

3.3. Comprehensive Evaluation of the Primary Model

3.3.1. Statistical Comparison: DeLong Tests

As Table 6 shows, the tuned model was statistically superior to every baseline: LightGBM ($p = 0.017$), XGBoost ($p = 0.005$), and by a larger margin against Random Forest, Logistic Regression, and MLP (all $p < 0.001$). The one non-significant result was against the untuned CatBoost baseline ($p = 0.619$), which is expected: hyperparameter tuning sharpened the model without fundamentally altering its architecture.

Table 6: DeLong test results for pairwise AUROC comparisons between tuned CatBoost and all baseline classifiers.

Comparison	AUROC (Tuned)	AUROC (Baseline)	p -value	Significant ($p < 0.05$)
vs CatBoost Baseline	0.9868	0.9863	0.619	No
vs LightGBM	0.9868	0.9806	0.017	Yes
vs XGBoost	0.9868	0.9780	0.005	Yes
vs Random Forest	0.9868	0.9656	< 0.001	Yes
vs Logistic Regression	0.9868	0.9152	< 0.001	Yes
vs MLP	0.9868	0.9212	< 0.001	Yes

3.3.2. Calibration

Pre-calibration, the model achieved a Brier Score of 0.0385, indicating strong probabilistic calibration. Platt scaling was applied as a post-hoc step; however, the calibrated model yielded a marginally higher Brier Score of 0.0422, a slight degradation rather than improvement, consistent with the model's decision boundary already being well-tuned through Optuna. The reliability diagram (Figure 6) shows that the model is well-calibrated in the intermediate probability range (0.3–0.8), with minor over-confidence at very high predicted probabilities. The pre-calibration model is used for all subsequent threshold analyses.

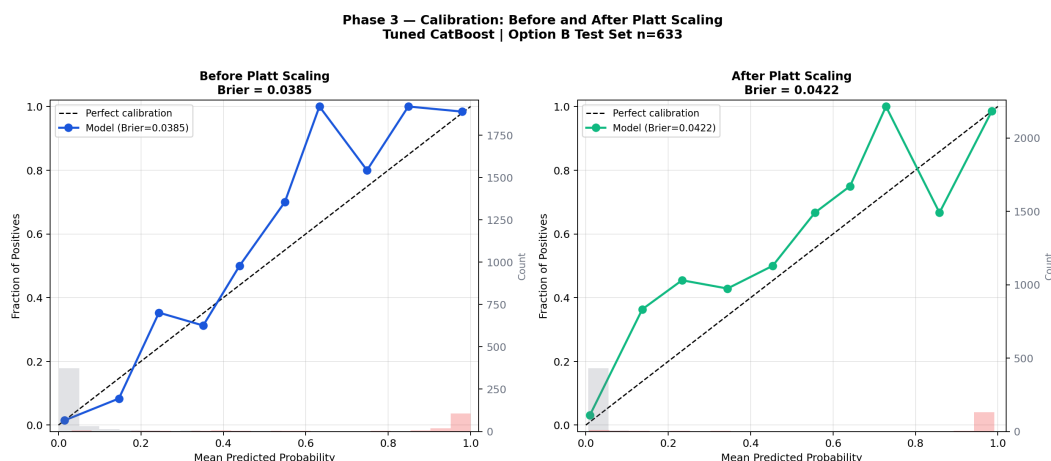


Figure 6: Reliability diagram (calibration curve) for the tuned CatBoost model before and after Platt scaling. Pre-calibration Brier Score: 0.0385; post-calibration: 0.0422. A perfectly calibrated model would follow the diagonal dashed line.

3.3.3. Decision Curve Analysis

The tuned CatBoost model demonstrated positive net benefit over both the *treat all* and *treat none* reference strategies across the clinically relevant threshold range of 0.1 to 0.8, confirming that deployment of the model would result in more appropriate interventions than either extreme clinical strategy (Figure 7).

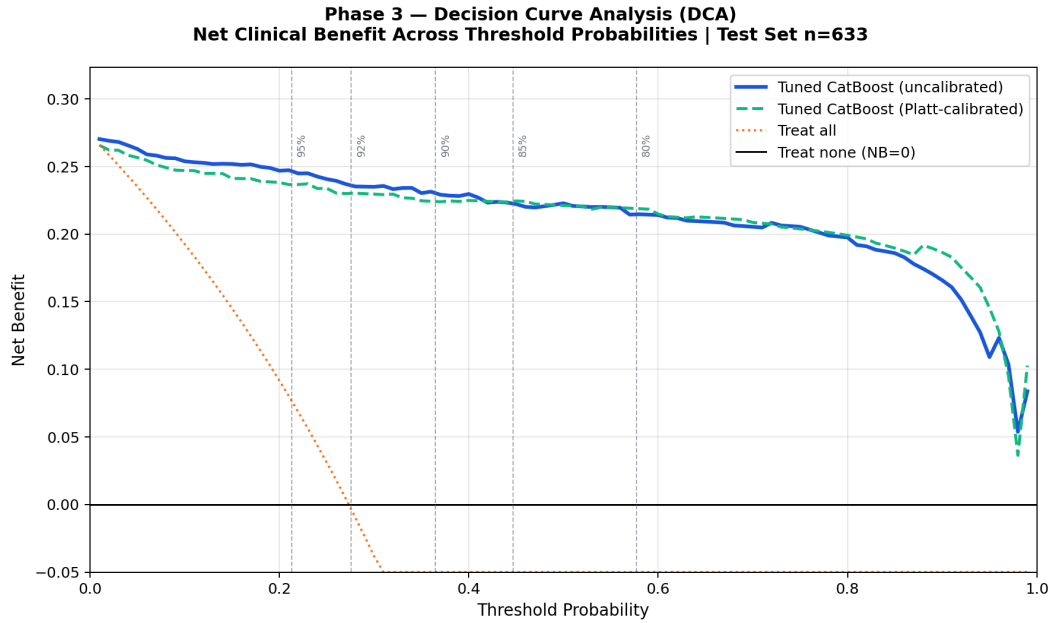


Figure 7: Decision Curve Analysis for the tuned CatBoost model. Net benefit is plotted against the decision threshold probability. The model outperforms both the treat-all and treat-none baselines across the range of clinically plausible thresholds.

3.3.4. Threshold Analysis and Clinical Operating Point

Table 7 presents five operating points spanning 80% to 95% sensitivity, with corresponding classification metrics at each decision boundary evaluated on the held-out test set.

Table 7: Threshold sensitivity analysis for the tuned CatBoost model on the Option B test set ($n = 633$). The $\approx 90\%$ sensitivity operating point ($t = 0.365$, achieved sensitivity 0.896) is recommended for clinical deployment. Sens. = sensitivity; Spec. = specificity.

Operating Point	Threshold	Sens.	Spec.	PPV	NPV	F1
80% Sensitivity	0.577	0.815	0.993	0.979	0.935	0.890
85% Sensitivity	0.447	0.861	0.980	0.943	0.949	0.900
$\approx 90\%$ (recommended)	0.365	0.896	0.963	0.901	0.961	0.899
92% Sensitivity	0.276	0.919	0.943	0.859	0.969	0.888
95% Sensitivity	0.213	0.954	0.930	0.838	0.982	0.892

A threshold of 0.365, targeting 90% sensitivity (achieved: 0.896), is recommended as the primary clinical operating point: sensitivity 0.896, specificity 0.963, PPV 0.901, NPV 0.961, F1 0.899. The high NPV means patients falling below the threshold can be safely deprioritised with a missed-sepsis rate of only 3.9%, clinically tolerable in a PICU context where routine monitoring continues regardless of model output, while the 96.3% specificity limits unnecessary escalations that would otherwise drive alert fatigue. Across the full threshold range, PPV drops from 0.979 to 0.838 moving from the 80% to 95% sensitivity point (the cost of capturing more true positives), while F1 stays stable (0.888–0.900) throughout, confirming the recommended 90% point sits near the F1 optimum while preserving workflow-compatible specificity.

3.3.5. Subgroup Analysis by Age Band

Results by age band are summarised in Table 8.

The model generalised robustly across all subgroups, with AUROC ranging from 0.970 in neonates to 1.000 in adolescents. The neonate group had the lowest prevalence (16.7%) and the most restrained performance, consistent

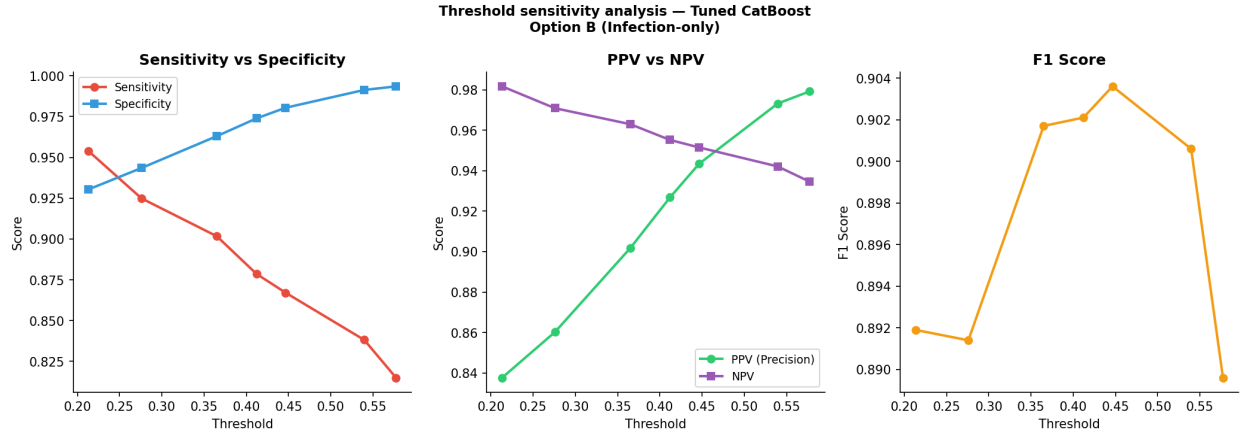


Figure 8: Threshold sensitivity curve showing sensitivity, specificity, PPV, and F1 score as a function of decision threshold. The recommended operating point ($t = 0.365$) is marked with a vertical dashed line.

Table 8: Subgroup performance by age band at the recommended threshold ($t = 0.365$). Prev. = sepsis prevalence within subgroup. The adolescent subgroup ($n = 19$) is small; perfect discrimination should be interpreted with caution.

Age Group	N	Sepsis N	Prev.	AUROC	Sens.	Spec.	F1	AUPRC
Neonate (0–28d)	102	17	16.7%	0.970	0.824	0.965	0.824	0.915
Infant (29d–1y)	223	64	28.7%	0.989	0.859	0.969	0.887	0.977
Child (1–12y)	289	87	30.1%	0.988	0.931	0.955	0.915	0.972
Adolescent (12–18y)	19	5	26.3%	1.000	1.000	1.000	1.000	1.000

with the known physiological atypicality of neonatal sepsis, though 0.970 remains strong relative to any published benchmark for this age group. The infant and child subgroups together constitute 80% of the test cohort (512 of 633 patients) and both exceeded AUROC 0.987, confirming discriminative performance is not concentrated in a single age stratum; the adolescent subgroup’s perfect discrimination on only 19 patients carries a wide confidence interval and does not warrant strong conclusions beyond an absence of evidence for degraded performance in older children. SHAP feature rankings were consistent across all age bands lactate, D-dimer, and platelets drive predictions throughout, with the neonate group showing slightly elevated reliance on haematological features.

3.4. Deep Learning Model Evaluation

Hyperparameter search converged to the configurations detailed in Tables 9–11. The FT-Transformer settled on a shallow two-block architecture with 256 embedding dimensions and eight attention heads, with relatively low dropout throughout, reflecting the moderate training set size. The ResNet required eight blocks with 512-dimensional hidden layers and substantially higher second-layer dropout (0.444), indicating that aggressive regularisation was needed to prevent overfitting in the deeper feed-forward stack. TabNet converged to four decision steps with 64-dimensional output width, a configuration that is common for clinical tabular datasets of this scale.

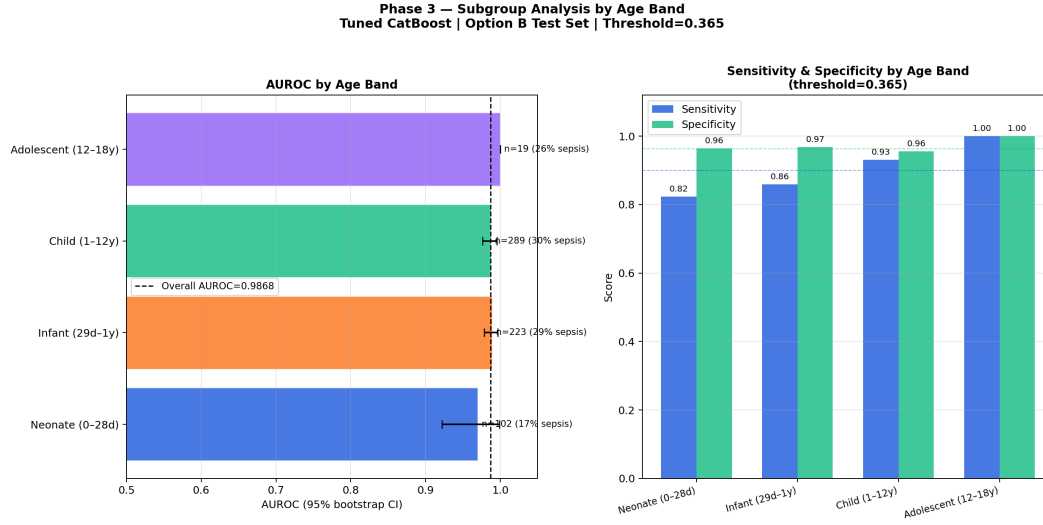


Figure 9: AUROC and sensitivity by age band with 95% confidence intervals. The neonate group shows the widest uncertainty interval attributable to its smaller sample size. All four age bands exceed AUROC 0.97.

Table 9: Optimal FT-Transformer hyperparameters identified by Optuna (best 5-fold CV AUPRC: 0.9453).

Hyperparameter	Value
Transformer blocks (n_blocks)	2
Block embedding dimension (d_block)	256
Attention heads	8
Attention dropout	0.142
FFN hidden multiplier	2.338
FFN dropout	0.066
Residual dropout	0.013
Learning rate	4.09×10^{-4}
Weight decay	7.71×10^{-5}
Batch size	128

Table 10: Optimal tabular ResNet hyperparameters identified by Optuna (best 5-fold CV AUPRC: 0.8594).

Hyperparameter	Value
Residual blocks (n_blocks)	8
Main hidden dimension (d_main)	512
Inter-block dimension (d_hidden)	512
First-layer dropout	0.204
Second-layer dropout	0.444
Learning rate	1.59×10^{-4}
Weight decay	1.67×10^{-6}
Batch size	64

Phase 3 — Evaluation Summary Dashboard
Tuned CatBoost | Option B (Infection-Only) | Test Set n=633

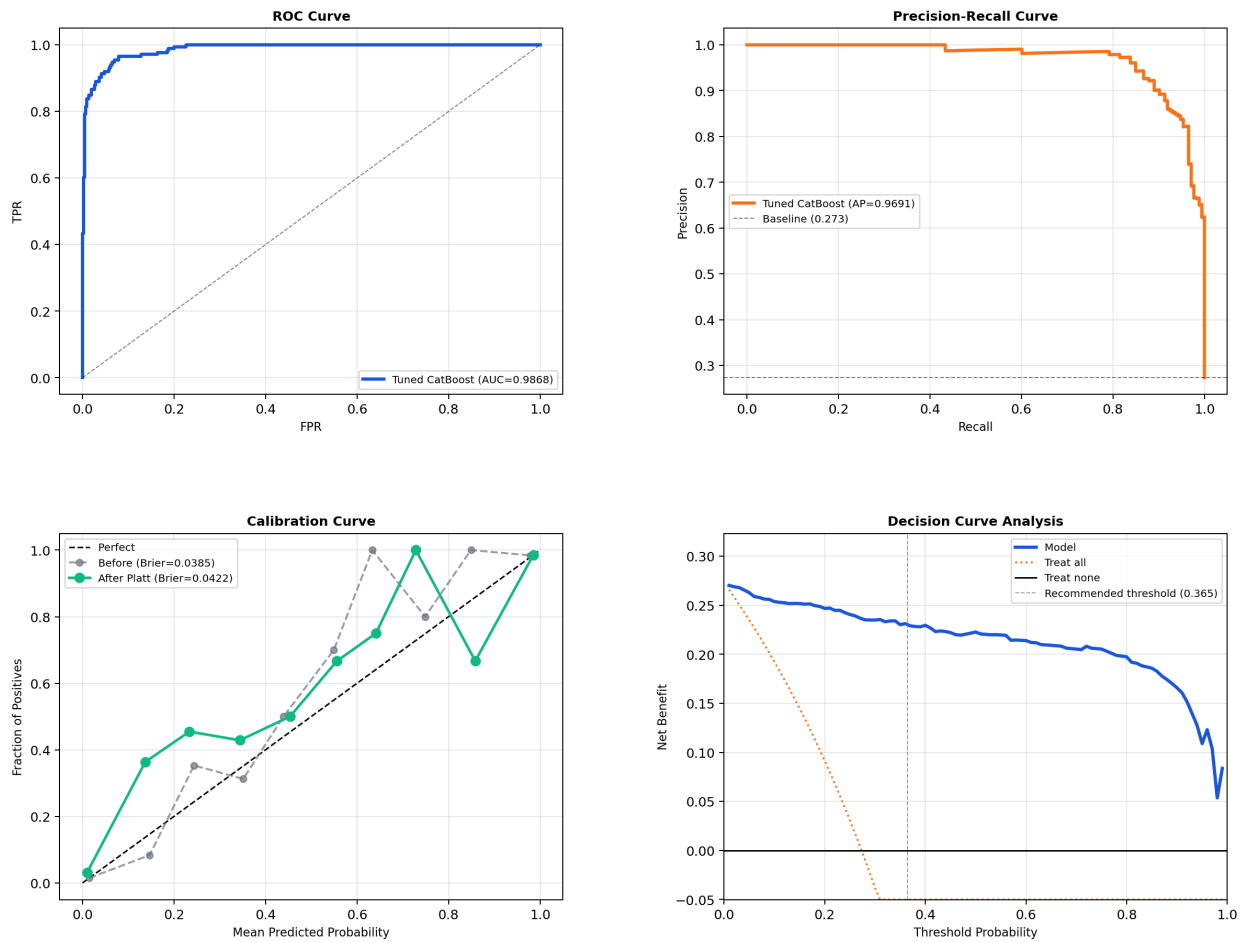


Figure 10: Comprehensive evaluation dashboard for the tuned CatBoost primary model, summarising ROC and Precision-Recall curves, calibration diagram, Decision Curve Analysis, confusion matrix, and subgroup AUROC results at the recommended threshold ($t = 0.365$).

Table 11: Optimal TabNet hyperparameters identified by Optuna (best 5-fold CV AUPRC: 0.9049).

Hyperparameter	Value
Decision step output width (n_d)	64
Attention embedding dimension (n_a)	16
Decision steps (n_{steps})	4
Feature reuse coefficient (γ)	1.082
Sparsity regularisation (λ_s)	2.41×10^{-3}
Learning rate	7.89×10^{-3}
Batch normalisation momentum	0.221
Attention mask type	entmax
Batch size	128
Virtual batch size	64

Test-set results for all three architectures, alongside the tuned CatBoost reference, are presented in Table 12. The FT-Transformer achieved the highest discriminative performance among the three deep learning models at AUROC 0.9577 (95% CI: 0.9404–0.9737) and AUPRC 0.9131 (95% CI: 0.8748–0.9484). ResNet and TabNet produced near-identical AUROC values of 0.9311 (95% CI: 0.9046–0.9538) and 0.9306 (95% CI: 0.9088–0.9497) respectively, with TabNet showing the weakest AUPRC at 0.8657. None of the three architectures approached the tuned CatBoost AUROC of 0.9868.

Table 12: Deep learning model performance on the Option B test set ($n = 633$) at the threshold targeting 90% sensitivity. DeLong p -values are pairwise comparisons against the tuned CatBoost primary model. 95% CIs computed by stratified bootstrap (1000 iterations).

Model	AUROC	AUPRC	Sens.	Spec.	F1	Brier
FT-Transformer	0.9577	0.9131	0.896	0.852	0.783	0.073
ResNet	0.9311	0.8744	0.902	0.813	0.752	0.094
TabNet	0.9306	0.8657	0.908	0.770	0.720	0.108
Tuned CatBoost (ref.)	0.9868	0.9691	0.896	0.963	0.899	0.0385

Pairwise DeLong tests against the primary model yielded $p = 0.169$ for the FT-Transformer, $p = 0.018$ for ResNet, and $p = 0.014$ for TabNet. The FT-Transformer result does not reach significance at the conventional 0.05 threshold, meaning its 0.029 AUROC gap relative to CatBoost cannot be statistically distinguished from sampling variation on a test set of 633 patients. The ResNet and TabNet gaps are statistically confirmed; both architectures are meaningfully inferior to the primary model in discriminative terms.

A more clinically consequential distinction emerges when specificity is examined alongside sensitivity. All three deep learning models achieve similar or higher sensitivity than CatBoost, but at substantially lower specificity. The FT-Transformer matches CatBoost’s sensitivity of 0.896 exactly, but specificity falls to 0.852 from CatBoost’s 0.963. ResNet and TabNet push sensitivity to 0.902 and 0.908 respectively, at the further cost of specificity dropping to 0.813 and 0.770. At a prevalence of 27.3%, this specificity gap translates directly into false-positive burden: at the same sensitivity level, the FT-Transformer generates approximately four times as many false alerts per true positive as CatBoost (computed from reported specificities at matched sensitivity 0.896: $FP/TP_{FT-T} \approx 0.44$ vs. $FP/TP_{CatBoost} \approx 0.11$). In a PICU deployment this matters operationally, because alert fatigue driven by low precision is one of the primary mechanisms through which clinical decision support systems erode clinician trust and compliance over time [21]. Probability calibration shows the same gradient: Brier scores range from 0.073 to 0.108 for the deep learning models compared to 0.0385 for tuned CatBoost, indicating that all three produce less reliable probability outputs as a basis for clinical threshold setting.

These results are broadly consistent with the tabular benchmark findings of Gorishniy et al., who showed that FT-Transformer becomes competitive with gradient boosting on datasets of sufficient scale, but that the advantage reverses at moderate training sizes where CatBoost’s implicit regularisation through ordered boosting provides more

stable generalisation than the data-hungry attention mechanism [12]. With 2,530 training patients, this cohort sits in that moderate-size regime. The TabNet outcome is informative in a specific way: instance-wise feature selection did not produce a performance gain relative to the simpler ResNet, suggesting that predictive information is distributed across the feature set broadly rather than concentrated in a sparse patient-specific subset. This has a clinical implication of its own, since it means no small panel of features is sufficient for reliable early sepsis detection uniformly across all patients in this cohort.

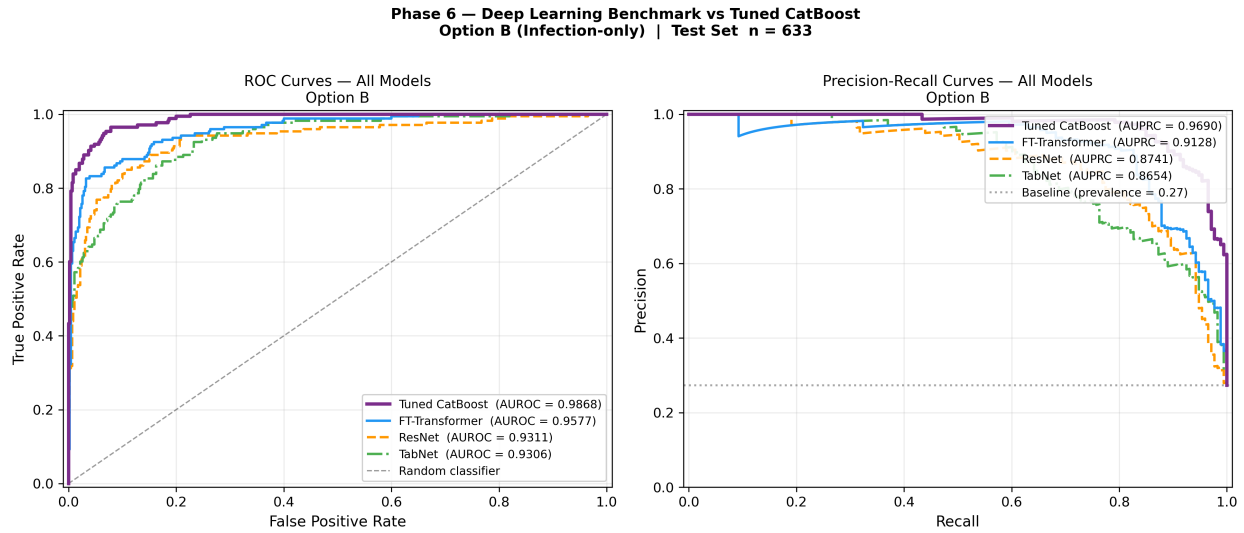


Figure 11: ROC (left) and Precision-Recall (right) curves for the three deep learning architectures compared against tuned CatBoost on the Option B test set. The FT-Transformer most closely approaches the CatBoost reference in both curves. ResNet and TabNet overlap substantially throughout the high-recall region.

3.5. CBFTTS-SA Model: CatBoost–FT-Transformer Stacking Ensemble

The real-data CBFTTS-SA model achieved AUROC 0.9851 (95% CI: 0.9770–0.9921), marginally below tuned CatBoost alone (0.9868); the hybrid (real + synthetic) CBFTTS-SA model configuration achieved AUROC 0.9786 (95% CI: 0.9683–0.9875). DeLong tests comparing each configuration against tuned CatBoost yielded $p = 0.388$ and $p = 0.126$ respectively; neither difference is statistically significant at the 0.05 threshold, so the two architectures are reported side by side in Table 13 and throughout this paper rather than as a strict ranking. Table 13 reports the full clinical evaluation for both CBFTTS-SA model configurations at the threshold targeting 90% sensitivity for each model, matching the protocol used for the deep learning benchmark (Section 3.4); SHAP explainability and the cohort-definition sensitivity check (Section 3.9) were performed on the CatBoost component only, since TreeSHAP and the cohort-definition question both apply most directly to the model recommended for immediate deployment; the weight analysis below shows CatBoost carries the large majority of the ensemble’s discriminative signal. To assess whether the hybrid (real + synthetic) CBFTTS-SA configuration is itself viable as a standalone deployment candidate rather than only as a statistically-tied alternative to CatBoost, Decision Curve Analysis, a full five-point threshold sensitivity analysis, and subgroup analysis by age band were additionally carried out for that configuration at its own recommended operating threshold; these extended results are reported in Section 3.5.1 below.

A secondary DeLong comparison against the standalone FT-Transformer (Section 3.4) shows the real-data CBFTTS-SA model performs significantly better than the transformer component alone ($p = 0.0013$); the hybrid CBFTTS-SA model’s advantage over the hybrid-trained FT-Transformer alone is directionally consistent but not significant ($p = 0.090$). This indicates that stacking meaningfully improves on the transformer component even though the ensemble does not surpass CatBoost alone, supporting the choice of FT-Transformer as the diversity-contributing base learner rather than confirming stacking as redundant.

Table 13: Tuned CatBoost versus the CBFTTS-SA model (CatBoost–FT-Transformer stacking ensemble) on the Option B test set ($n = 633$), at the threshold targeting 90% sensitivity for each model. 95% CIs for AUROC and AUPRC computed by stratified bootstrap (1000 iterations); see main text.

Configuration	AUROC	AUPRC	Sens.	Spec.	F1	Brier	DeLong p
Tuned CatBoost (single model)	0.9868	0.9691	0.896	0.963	0.899	0.0385	Reference
CBFTTS-SA model (real data)	0.9851	0.9658	0.902	0.957	0.894	0.0428	0.388
CBFTTS-SA model (hybrid, real+synthetic)	0.9786	0.9534	0.908	0.935	0.872	0.0503	0.126

The meta-learner weight ratio assigned by logistic regression to the two base learners is informative: CatBoost received approximately three to four times the weight of the FT-Transformer across both configurations, confirming that at the present cohort size the ensemble’s predictive signal is driven predominantly by the gradient-boosted component. This is consistent with the benchmark finding in Section 3.4 that FT-Transformer-class architectures close the gap with gradient boosting only as training data scales [12]; at $n = 2,530$ training patients, CatBoost’s ordered-boosting regularisation still has the edge. The CBFTTS-SA model is nonetheless presented as this study’s featured architecture (Section 1), on the basis that it is statistically indistinguishable from CatBoost alone today and is architecturally the more promising candidate to improve further as larger paediatric ICU datasets become available, whereas tuned CatBoost alone remains the stronger-performing and fully interpretable choice for immediate deployment at the present cohort size. The hybrid CBFTTS-SA model configuration produced a slightly lower AUROC than the real-data configuration, consistent with the finding in Section 3.7 that synthetic patients introduce noise the sample weighting scheme does not fully suppress.

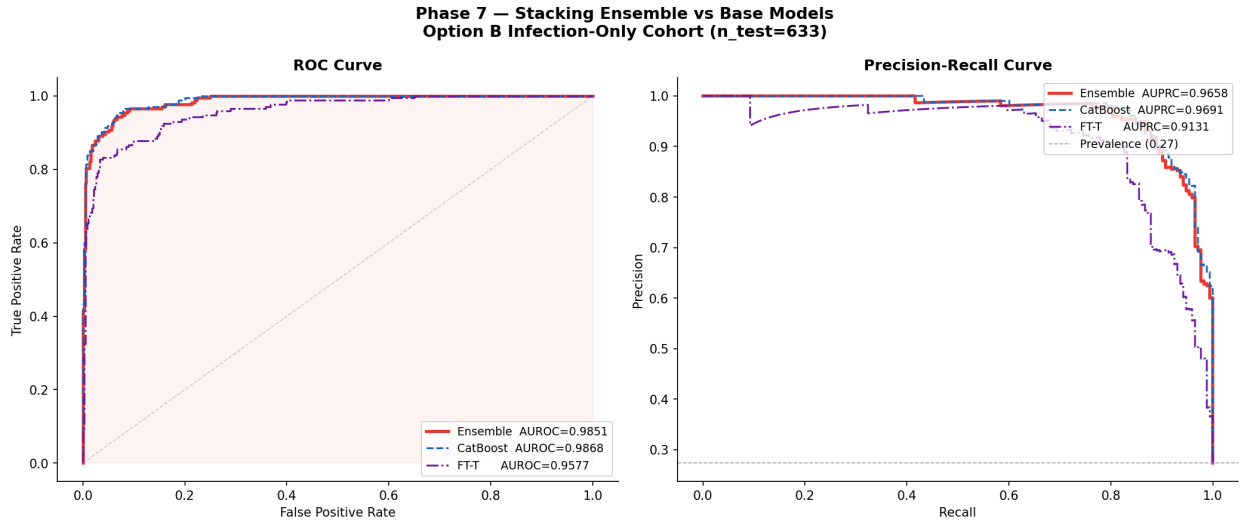


Figure 12: ROC (left) and Precision-Recall (right) curves for both CBFTTS-SA model configurations compared against tuned CatBoost on the Option B test set.

3.5.1. Extended Clinical Evaluation of the Hybrid CBFTTS-SA Model

The evaluation above establishes that the hybrid CBFTTS-SA model is statistically indistinguishable from tuned CatBoost at a single operating point targeting 90% sensitivity (Table 13). To characterise this configuration more completely as a candidate for future deployment as paediatric ICU datasets scale, the same threshold-sensitivity, subgroup, and net-benefit analyses applied to the primary CatBoost model in Sections 3.3.4 and 3.3.5 were repeated for the hybrid CBFTTS-SA model in isolation, using its own recommended operating threshold of $t = 0.233$ (targeting $\approx 90\%$ sensitivity, achieved sensitivity 0.9075) rather than borrowing the CatBoost threshold of $t = 0.365$, since the

two models' predicted-probability distributions are not identical and applying a single model's threshold to another would misrepresent its true operating characteristics.

Threshold sensitivity. Table 14 presents the same five operating points spanning 80–95% sensitivity used for the primary model. The pattern mirrors the CatBoost result: specificity trades off against sensitivity from 0.978 at the 80% point down to 0.852 at the 95% point, while F1 remains comparatively stable (0.813–0.885) across the range, peaking at the 85% operating point (F1 = 0.885) rather than exactly at the 90% point, though the difference between the two (0.885 vs 0.872) is marginal.

Table 14: Threshold sensitivity analysis for the hybrid (real + synthetic) CBFTTS-SA model on the Option B test set ($n = 633$). The $\approx 90\%$ sensitivity operating point ($t = 0.233$, achieved sensitivity 0.908) is used as the model's own recommended threshold throughout this subsection. Sens. = sensitivity; Spec. = specificity.

Operating Point	Threshold	Sens.	Spec.	PPV	NPV	F1
80% Sensitivity	0.605	0.804	0.978	0.933	0.930	0.863
85% Sensitivity	0.402	0.867	0.965	0.904	0.951	0.885
$\approx 90\%$ (recommended)	0.233	0.908	0.935	0.840	0.964	0.872
92% Sensitivity	0.184	0.925	0.926	0.825	0.970	0.872
95% Sensitivity	0.045	0.954	0.852	0.708	0.980	0.813

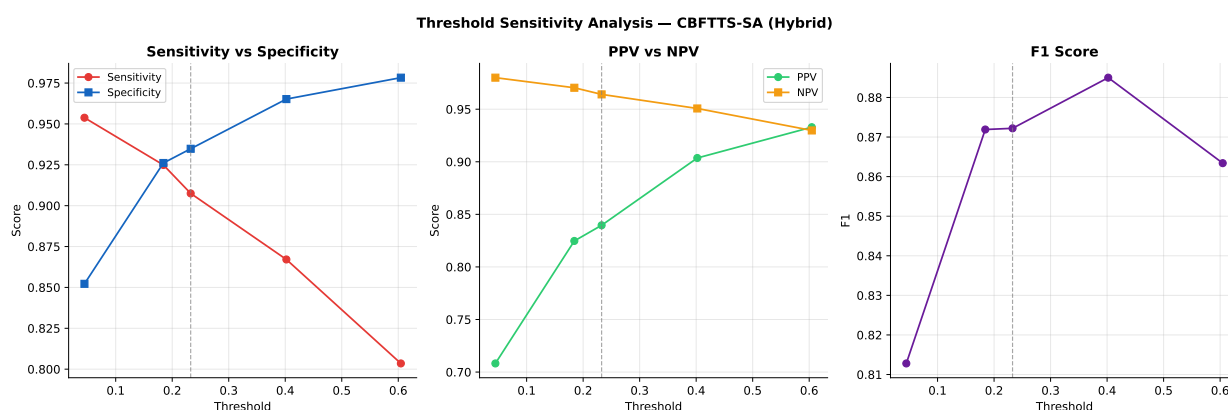


Figure 13: Threshold sensitivity curves for the hybrid CBFTTS-SA model showing sensitivity, specificity, PPV, NPV, and F1 score as a function of decision threshold. The recommended operating point ($t = 0.233$) is marked with a vertical dashed line.

Subgroup analysis by age band. Table 15 reports performance by age band at the model's recommended threshold. AUROC ranges from 0.961 in neonates to 1.000 in adolescents, closely tracking the CatBoost subgroup pattern (0.970–1.000, Table 8) though 0.5–0.9 percentage points lower in every band except the adolescent group, consistent with the small but consistent AUROC gap between the hybrid CBFTTS-SA model and CatBoost observed at the overall cohort level (Table 13). As with the primary model, the neonate subgroup shows the most restrained performance and the smallest prevalence (16.7%), and the infant and child subgroups together account for 80% of the test cohort, confirming that the hybrid model's discriminative performance is likewise not concentrated in a single age stratum.

Decision Curve Analysis. The hybrid CBFTTS-SA model demonstrated positive net benefit over both the *treat all* and *treat none* reference strategies across the clinically relevant threshold range of 0.1 to 0.8, the same range used for the primary CatBoost model's DCA (Section 3.3.4), and in fact remained positive across the entire threshold range tested (0.01–0.95; Figure 15), confirming that the hybrid configuration would, like CatBoost, translate into more appropriate clinical decisions than either extreme strategy if deployed. Net benefit declines gradually and stays close to zero from 0.266 at a threshold of 0.01 down to 0.058 at 0.95, in contrast with the treat-all strategy, whose net benefit

Table 15: Subgroup performance by age band for the hybrid (real + synthetic) CBFTTS-SA model, at its recommended threshold ($t = 0.233$). Prev. = sepsis prevalence within subgroup. As with the primary model, the adolescent subgroup ($n = 19$) is small and perfect discrimination should be interpreted with caution.

Age Group	N	Sepsis N	Prev.	AUROC	Sens.	Spec.	F1	AUPRC
Neonate (0–28d)	102	17	16.7%	0.961	0.765	0.965	0.788	0.871
Infant (29d–1y)	223	64	28.7%	0.982	0.859	0.962	0.880	0.962
Child (1–12y)	289	87	30.1%	0.983	0.966	0.901	0.880	0.963
Adolescent (12–18y)	19	5	26.3%	1.000	1.000	0.929	0.909	1.000

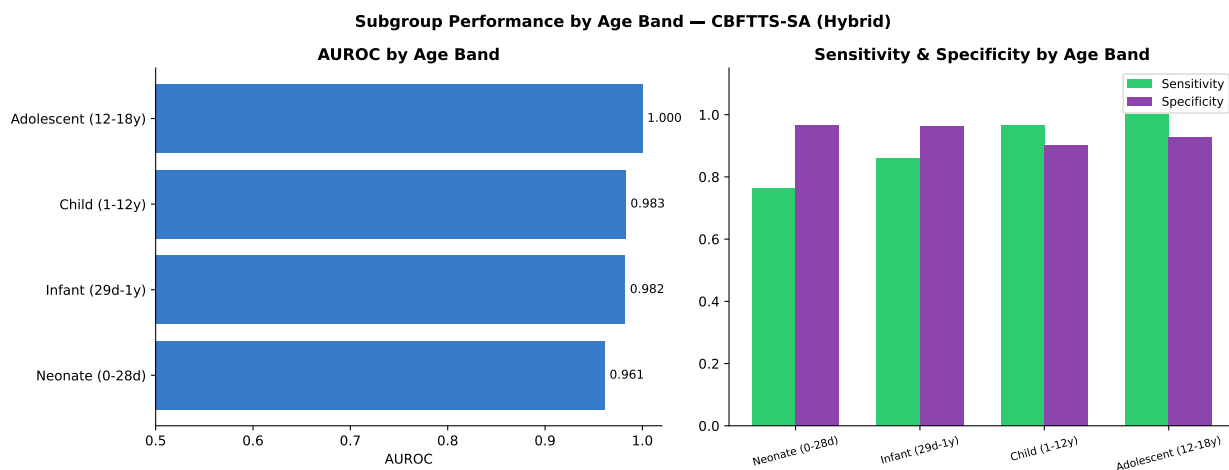


Figure 14: AUROC, sensitivity, and specificity by age band for the hybrid CBFTTS-SA model at its recommended threshold ($t = 0.233$). All four age bands exceed AUROC 0.96, mirroring the age-band pattern observed for the primary CatBoost model (Figure 9).

collapses once the decision threshold exceeds the cohort prevalence of 27.3% (treat-all net benefit crosses zero at approximately threshold 0.27 and falls sharply thereafter, reaching approximately -13.5 at threshold 0.95). Figure 15 plots the full 0.01–0.95 range on a single axis, so this divergence dominates the vertical scale and compresses the model curve into a comparatively flat band near zero; the substantive comparison is nonetheless unambiguous, since the model curve never crosses the treat-all curve below the prevalence threshold and remains above the treat-none baseline (NB = 0) throughout, indicating a consistent clinical advantage across the plausible range of deployment thresholds rather than only at the single 90%-sensitivity operating point emphasised elsewhere in this paper.

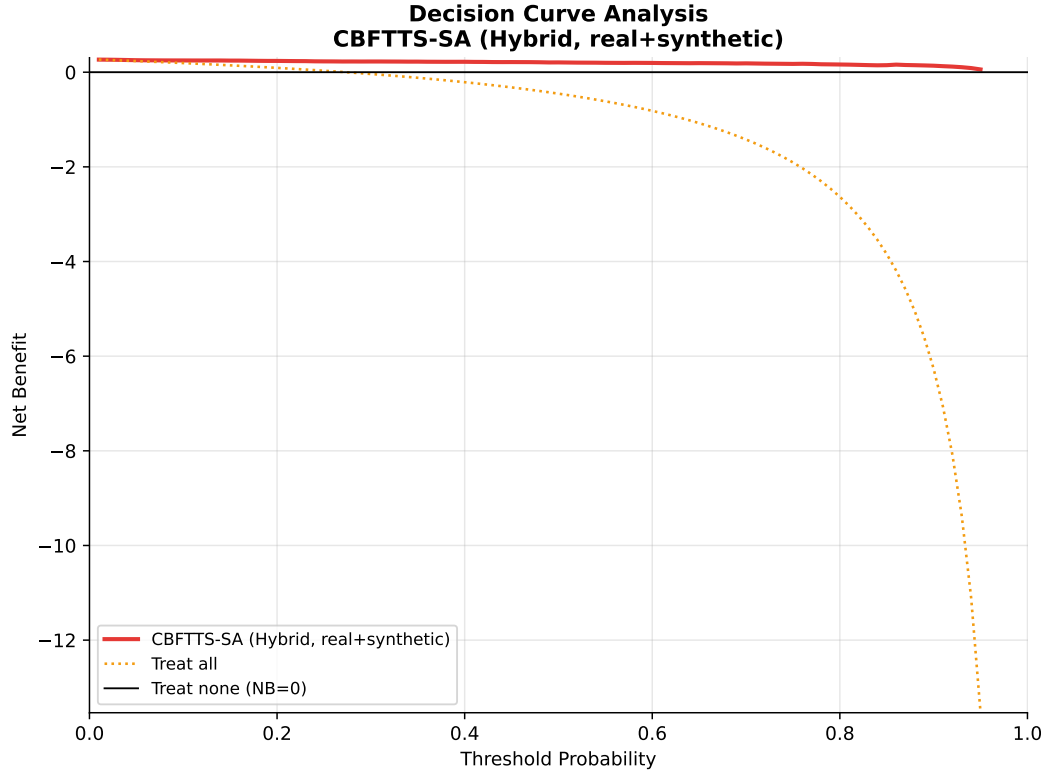


Figure 15: Decision Curve Analysis for the hybrid CBFTTS-SA model. Net benefit is plotted against the decision threshold probability over the full 0.01–0.95 range tested, alongside the treat-all and treat-none reference strategies. Because the treat-all curve falls sharply below the cohort’s 27.3% prevalence threshold (reaching approximately -13.5 at threshold 0.95), the axis is dominated by that divergence and the model’s own curve appears as a comparatively flat band close to zero; see main text for the corresponding CatBoost comparison plotted on a clinically-windowed axis (Figure 7).

Taken together, these extended results indicate that the hybrid CBFTTS-SA model’s near-parity with CatBoost on overall AUROC (Table 13) extends to the clinically relevant threshold-sensitivity, subgroup, and net-benefit profile: the hybrid configuration is a coherent, self-consistent clinical tool in its own right rather than only a statistically-tied summary statistic, which supports its designation as this study’s forward-looking architecture (Section 1) even though tuned CatBoost remains the stronger and more directly interpretable choice for deployment at the present cohort size.

3.6. Attention-Augmented CatBoost

The result was a statistically significant degradation in performance. The attention-augmented model achieved AUROC 0.9657 on the test set, compared to 0.9868 for tuned CatBoost alone. A pairwise DeLong test confirmed the gap was not attributable to sampling variation: $p = 0.014$, with the augmented model performing significantly worse. The Brier score rose from 0.0385 to 0.0978 alongside the discriminative decline, indicating that probability calibration deteriorated in parallel.

Examining the features assigned the highest attention mass by the FT-Transformer reveals why augmentation failed rather than helped. The five most attention-weighted features were `calcium_max`, `pao2_max`, care unit designation (SICU), `lactate_max`, and `platelets_min`. The final two are among the top five features by mean absolute SHAP value in the CatBoost model, and the others are strong physiological predictors in their own right: the attention weights were not identifying novel information, but reproducing, in a noisier continuous form, the same importance ranking CatBoost’s split-based mechanism already encodes. Adding these 225 attention columns increased feature dimensionality without adding predictive content.

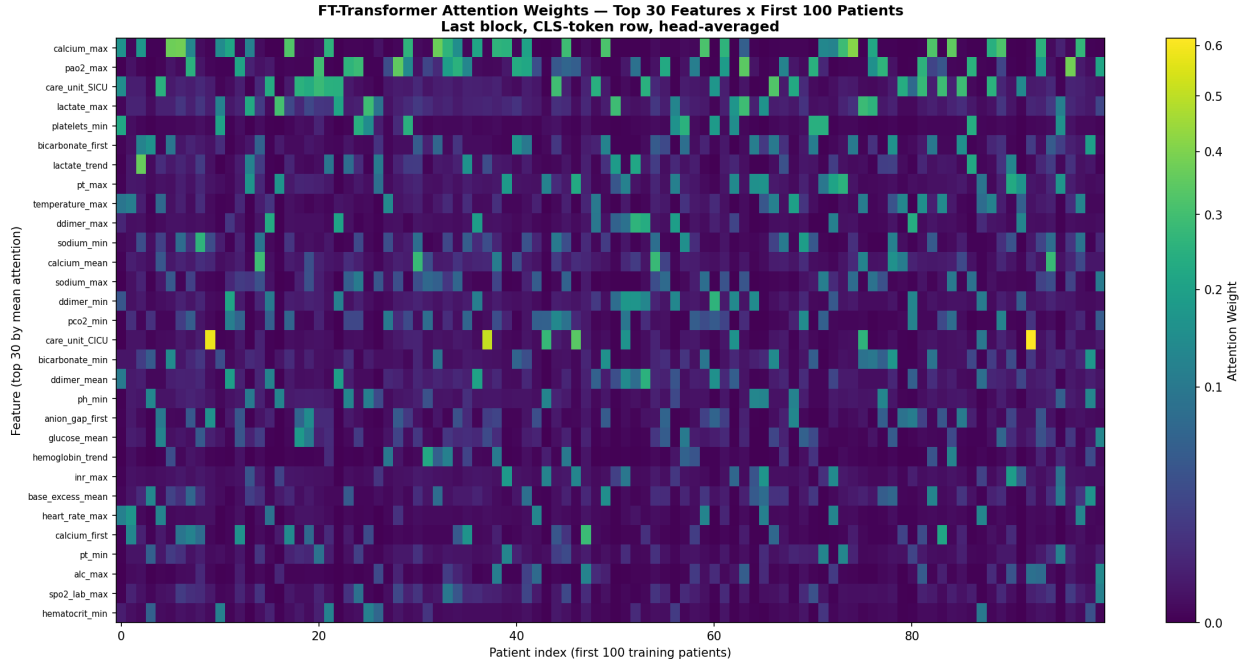


Figure 16: FT-Transformer CLS-token attention weight distribution across the 225 input features. Attention mass concentrates on `calcium_max`, `pao2_max`, `lactate_max`, and `platelets_min` features that are already among the strongest predictors in the CatBoost model.

This outcome is a useful negative result: it suggests that the FT-Transformer and CatBoost, despite their architectural differences, draw on substantially overlapping predictive structure within this feature set, so injecting the transformer’s summary of that structure into CatBoost as additional features adds correlated inputs without a corresponding gain in discriminative signal [12]. This is a distinct finding from the CBFTTS-SA model result above: stacking the two architectures’ independent predictions (Section 3.5) is statistically neutral, whereas concatenating the FT-Transformer’s internal representations directly into CatBoost’s input space is actively harmful.

3.7. Synthetic Data Augmentation

As a diagnostic step, vanilla CTGAN was first applied to the full training dataset before any correction. The result exposed a severe limitation: the Spearman correlation between INR and PT in real training data (0.999) collapsed to 0.016 after vanilla CTGAN generation, meaning a synthetic patient could exhibit a normal INR alongside a massively prolonged PT a physiologically impossible combination. The three-stage correction pipeline (Section 2.2.7) substantially repaired this correlation structure.

Three augmentation configurations were evaluated against the real-data tuned CatBoost model (Table 16). A CatBoost model trained exclusively on the 10,000 synthetic patients achieved AUROC 0.9290 ($p < 0.0001$, significantly worse). A hybrid CatBoost combining all 2,530 real patients (weight 4) with the synthetic patients (weight 1) improved to AUROC 0.9776 ($p = 0.0007$, still significantly worse). The hybrid CBFTTS-SA model configuration, in which the hybrid dataset trained the base learners before the meta-learner combined them, reached AUROC

0.9786 the only augmentation configuration whose DeLong test did not reach significance ($p = 0.126$), because the meta-learner’s dominant real-data CatBoost weighting limits how much synthetic noise reaches the final output.

Table 16: Synthetic data augmentation results. All DeLong p -values compare against the real-data tuned CatBoost model (AUROC 0.9868).

Configuration	AUROC	DeLong p	Significant?
Synthetic-only CatBoost	0.9290	< 0.0001	Yes (worse)
Hybrid CatBoost (weighted)	0.9776	0.0007	Yes (worse)
★ Hybrid CBFTTS-SA model (Proposed)	0.9786	0.126	No
Tuned CatBoost (real data)	0.9868		Reference

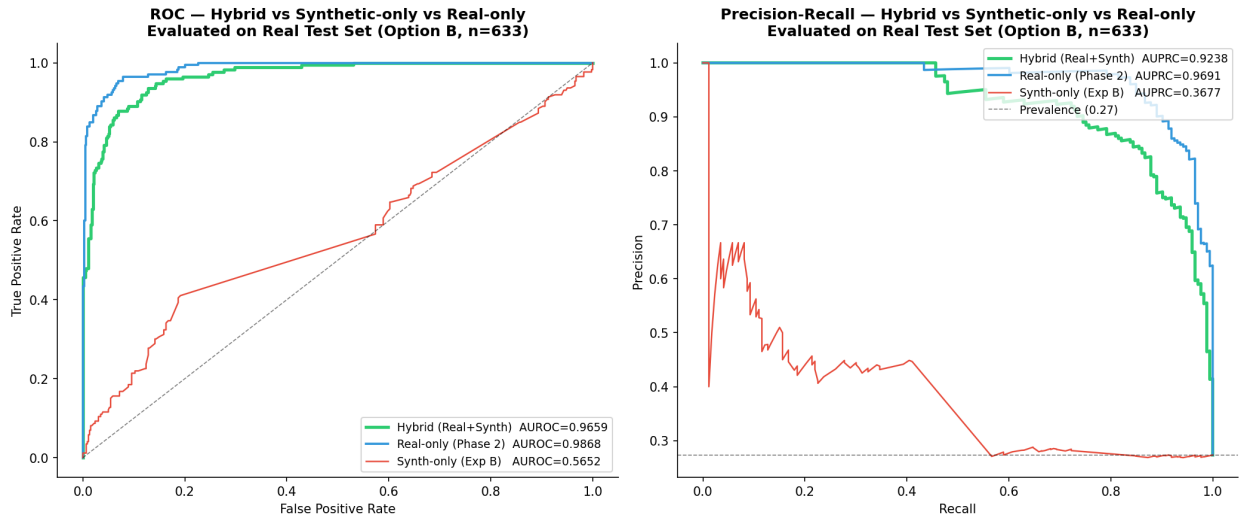


Figure 17: Three-way ROC and Precision-Recall comparison between the synthetic-only, hybrid-weighted, and real-data CatBoost configurations on the Option B test set.

The clear pattern across all three configurations is monotonic: AUROC climbs from 0.929 (pure synthetic) to 0.978 (hybrid, real-weighted) to 0.986 (real data only) as the real-data contribution to training increases. This is evidence that real data, not synthetic data, is doing the work synthetic patients act as a diluent rather than a booster. The underlying limitation is not pairwise correlation structure, which the Iman-Conover step restores reasonably well, but the long-tail distributions of abnormal laboratory values (e.g. lactate_max, D-dimer_max) that carry the strongest sepsis signal: a generative model trained on a few hundred positive examples systematically underestimates probability mass in the extreme right tail, and no post-hoc correlation correction can recover information the training data never provided. The practical implication is that synthetic augmentation is not a substitute for real data collection in this setting.

3.8. SHAP Explainability Analysis

SHAP explainability analysis was applied to the CatBoost component, which the weight analysis in Section 3.5 shows carries the large majority of the CBFTTS-SA model’s discriminative signal.

Global feature importance.. Global feature importance was ranked by mean absolute SHAP value across all test patients, reflecting each feature’s average contribution to predictions regardless of direction. The top 20 features are presented in Table 17, visualised as a beeswarm plot in Figure 19.

The top five features lactate, D-dimer, platelets, INR, and PT map directly onto the two computable Phoenix domains: elevated maximum lactate and depressed minimum platelets correspond to the cardiovascular domain, while

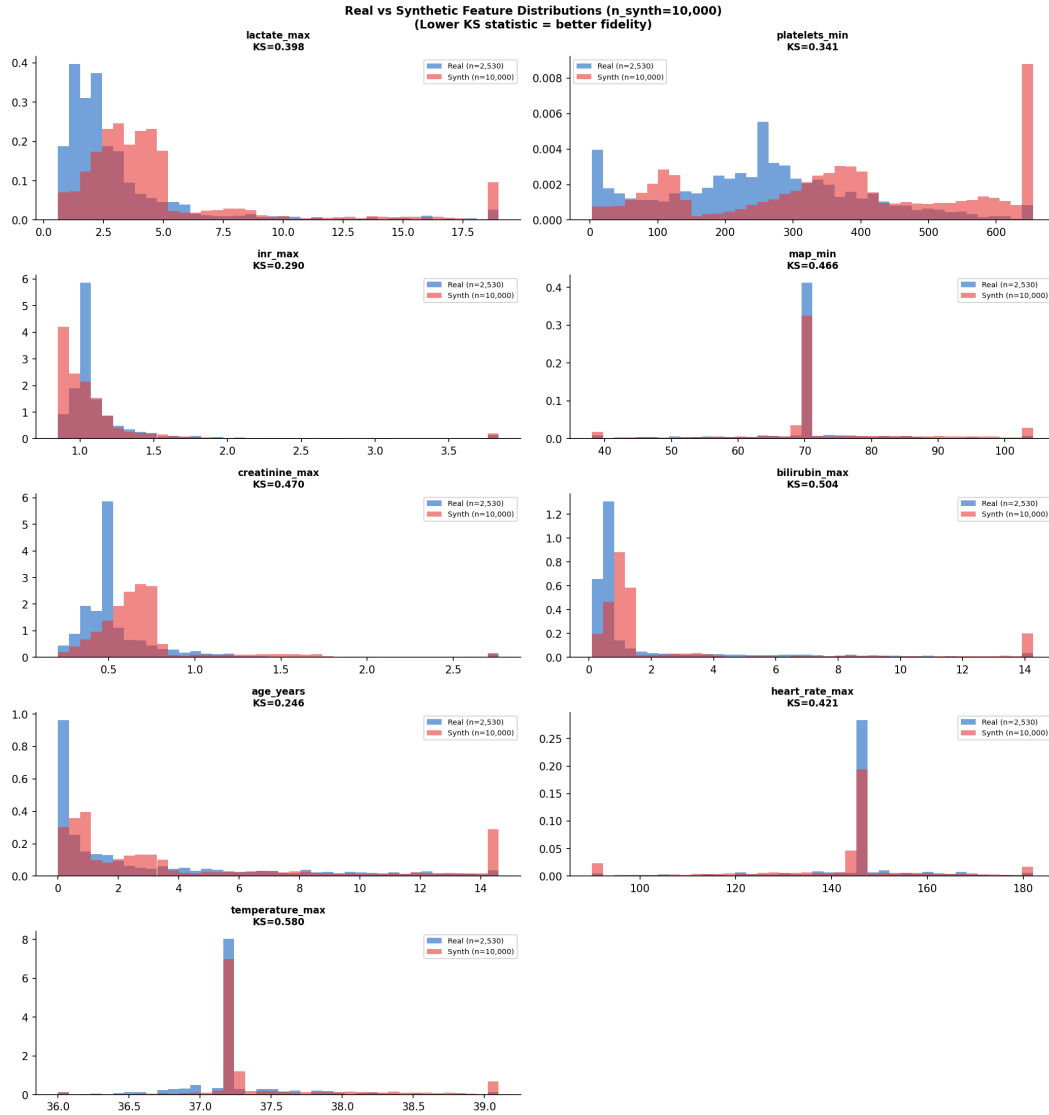


Figure 18: Comparison of key feature distributions between real training data and CTGAN-generated synthetic data after Iman-Conover correlation injection. Marginal distributions are broadly preserved, but the extreme right tails of `lactate_max` and `D-dimer_max` are underrepresented in the synthetic cohort.

Table 17: Top 20 features by mean absolute SHAP value, computed across all 633 test-set patients. Rankings reflect global importance regardless of prediction direction.

Rank	Feature	Display Name	Mean SHAP
1	<code>lactate_max</code>	Lactate (max)	0.992
2	<code>ddimer_max</code>	D-dimer (max)	0.957
3	<code>platelets_min</code>	Platelets (min)	0.776
4	<code>inr_max</code>	INR (max)	0.539
5	<code>calcium_max</code>	Calcium (max)	0.519
6	<code>pt_max</code>	PT (max)	0.259
7	<code>calcium_mean</code>	Calcium (mean)	0.222
8	<code>glucose_max</code>	Glucose (max)	0.214
9	<code>pao2_max</code>	PaO ₂ (max)	0.176
10	<code>ph_min</code>	pH (min)	0.161
11	<code>hematocrit_min</code>	Haematocrit (min)	0.146
12	<code>platelets_first</code>	Platelets (first recorded)	0.146
13	<code>care_unit_SICU</code>	Care unit: SICU	0.124
14	<code>fibrinogen_min</code>	Fibrinogen (min)	0.115
15	<code>care_unit_CICU</code>	Care unit: CICU	0.114
16	<code>ddimer_mean</code>	D-dimer (mean)	0.108
17	<code>platelets_mean</code>	Platelets (mean)	0.091
18	<code>potassium_min</code>	Potassium (min)	0.084
19	<code>lactate_first</code>	Lactate (first recorded)	0.075
20	<code>bicarbonate_min</code>	Bicarbonate (min)	0.073

elevated INR, D-dimer, and PT correspond to the coagulation domain [3, 2]. Calcium (max and mean) at ranks 5 and 7 reflects dysregulated electrolyte balance in severe systemic infection, and PaO₂, pH, and bicarbonate in the mid-tier capture respiratory and metabolic acidosis. Prior SHAP analyses on the PIC database have surfaced similarly coherent feature hierarchies [23, 24].

SHAP dependence analysis. For `lactate_max`, the SHAP contribution is approximately zero below 2 mmol/L and rises sharply above 4 mmol/L, consistent with the clinical threshold for hyperlactataemia in the Phoenix cardiovascular domain [3]. D-dimer shows a steep positive contribution above 2 mg/L (coagulopathy burden), platelets (min) shows a strongly negative contribution below 100×10^9 /L (thrombocytopenia), and INR contributions rise sharply above 1.5. Calcium (max) is non-monotonic: both hypo- and hypercalcaemia are associated with increased sepsis risk, a U-shaped pattern that cannot be recovered by linear models and that gives a mechanistic reason for the AUROC gap between gradient boosting and logistic regression in the baseline comparison.

Individual patient explanations. Three cases illustrate the model’s behaviour at its margins (Figure 21). The true positive (predicted probability 0.9997) shows elevated lactate, high D-dimer, and depressed platelets acting in concert. The false negative (probability 0.365, at threshold) reflects a borderline case: modest lactate elevation counterbalanced by near-normal coagulation markers, characteristic of early or atypical sepsis. The false positive (probability 0.982; true label non-sepsis) shows laboratory findings mimicking sepsis, possibly a severe non-infectious inflammatory state. The model’s errors are concentrated at the genuine physiological boundary of the Phoenix criteria, not in clearly normal or abnormal presentations.

Feature interaction analysis. The lactate–platelets interaction is the largest pairwise SHAP interaction in the model (mean absolute interaction 0.035), spanning the cardiovascular Phoenix domain: tissue hypoperfusion and consumptive thrombocytopenia frequently co-occur in the same haemodynamic cascade [3]. The INR–D-dimer coagulopathy axis is second largest (0.021), reflecting converging evidence of clotting factor depletion and fibrin degradation [2].

SHAP summary and clinical interpretation. The dominant features lactate, D-dimer, platelets, INR, and PT collectively represent three of the four Phoenix organ systems: cardiovascular (lactate, platelets), coagulation (D-dimer,

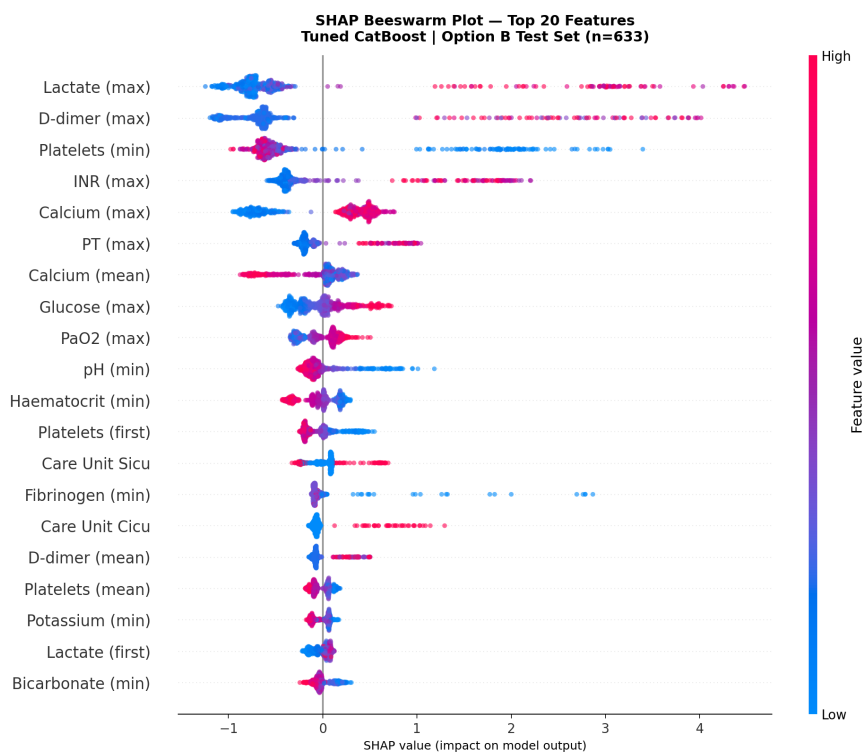


Figure 19: SHAP beeswarm plot for the top 20 features by mean absolute SHAP value, computed across all 633 test-set patients. Each point represents one patient; colour encodes feature value (red = high, blue = low). Features mapping to the Phoenix cardiovascular domain (lactate_max, platelets_min) occupy the top ranks, followed by the coagulation axis (ddimer_max, inr_max, pt_max).

SHAP Dependence Plots — Top 5 Clinical Features
Tuned CatBoost | Option B Test Set (n=633)
Colour = feature value (red=high, blue=low)

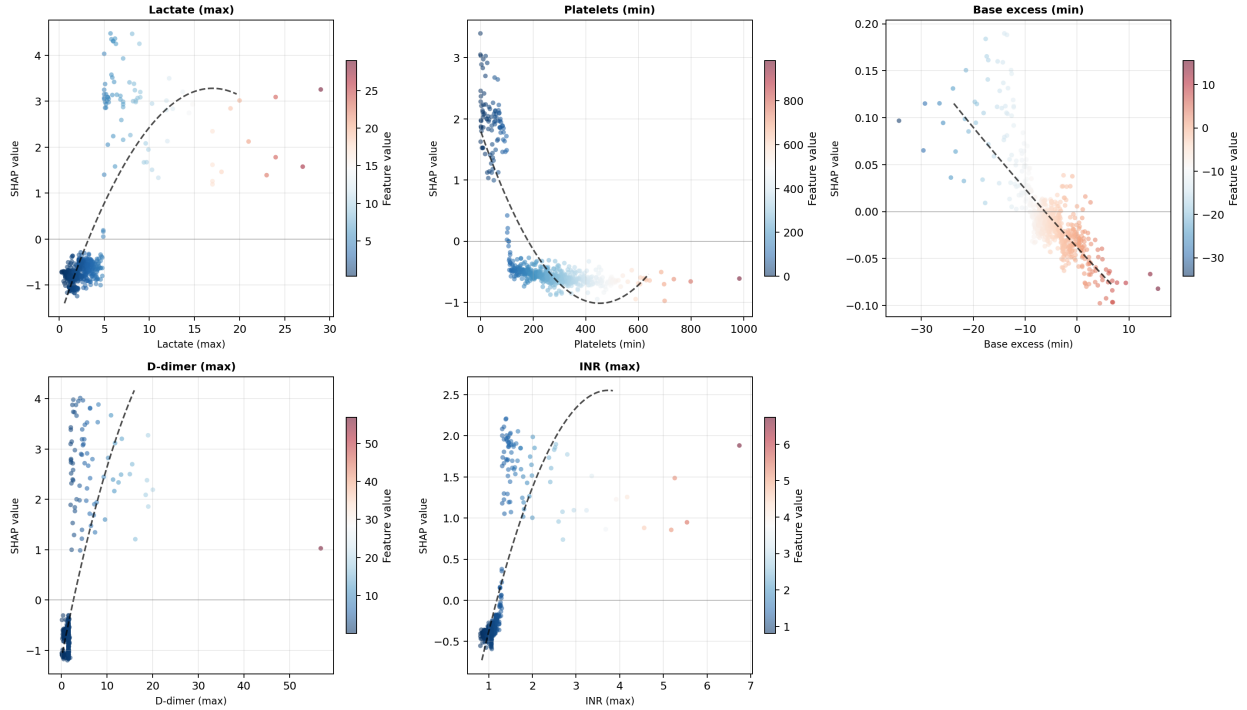


Figure 20: SHAP dependence plots for the top features. The x-axis shows raw feature value; the y-axis shows SHAP contribution to the sepsis prediction. Colour encodes the value of the highest-interacting feature identified by SHAP. The non-monotonic calcium pattern and the threshold-like behaviour of lactate and D-dimer are visible in the respective panels.

SHAP Waterfall Plots — Individual Patient Explanations
Tuned CatBoost | Option B Test Set | Threshold=0.365

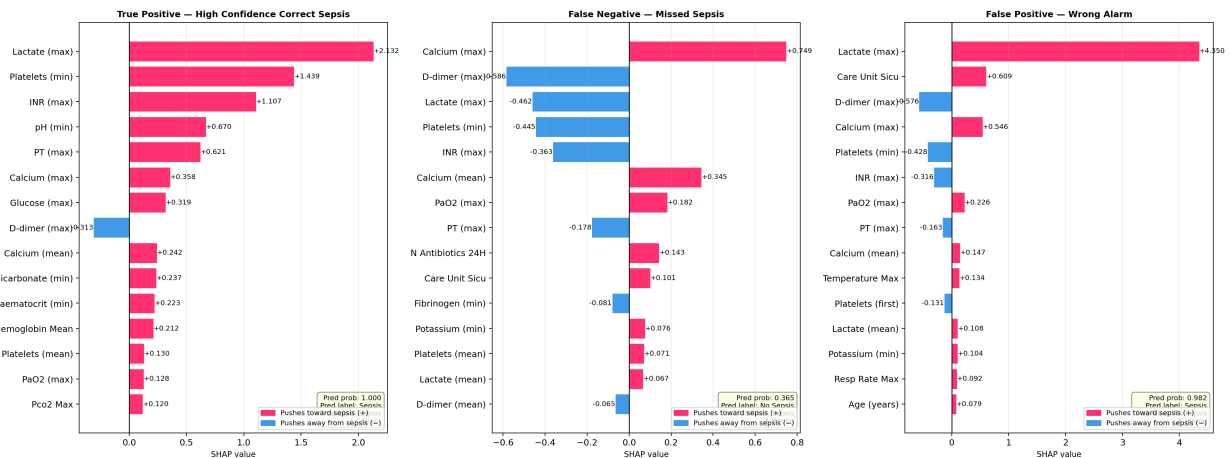


Figure 21: SHAP waterfall plots for three selected test patients: true positive (top), false negative (middle), and false positive (bottom). Red bars push the prediction toward the sepsis class; blue bars push toward non-sepsis. $f(x)$ is the final model output and $E[f(x)]$ is the baseline log-odds value.

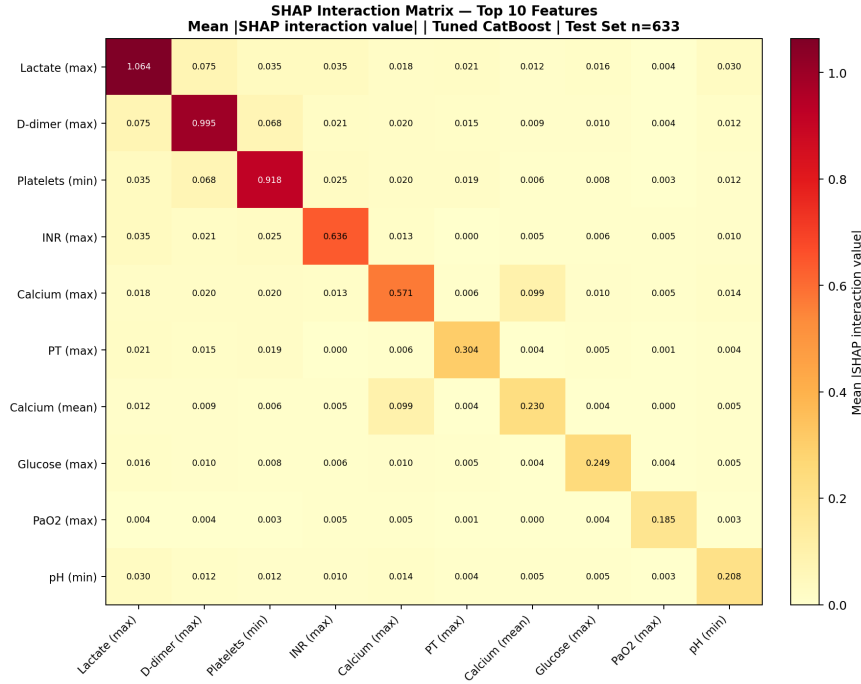


Figure 22: SHAP interaction value heatmap for the top features. The lactate–platelets axis (mean absolute interaction 0.035) and the INR–D-dimer coagulopathy axis (0.021) are the two dominant pairwise interactions, mapping to the Phoenix cardiovascular and coagulation domains respectively.

INR, PT), and metabolic (glucose, pH, bicarbonate). The absence of respiratory and neurological features from the top rankings is consistent with the known data limitation: FiO₂ and GCS were unavailable in PIC v1.1.0 and were set to zero in the Phoenix score computation [8]. The interaction analysis shows the model has internally learned that simultaneous multi-system compromise is disproportionately informative, consistent with the additive organ dysfunction logic underlying the Phoenix score itself, providing a principled basis for clinician trust in the model’s decision logic [21]. Age does not rank highly among the top-ranked features, suggesting organ dysfunction markers are sufficient discriminators across the paediatric age range without requiring demographic adjustment [3].

3.9. Sensitivity Analysis: Option A vs Option B Cohort

The Option A training set comprised 9,957 patients with a sepsis prevalence of 6.9%, compared to 2,530 patients at 27.3% for Option B. Performance on the same held-out test set is compared in Table 18.

Option A achieved a marginally higher AUROC (0.9968 vs 0.9868) and lower Brier Score (0.0164 vs 0.0385), both attributable to its larger training set and to the relative ease of discriminating sepsis from the broader non-infected majority: when 74.6% of the evaluation population has no infection suspicion at all, many true negatives are clinically straightforward and easy to classify correctly, inflating discriminative and calibration metrics beyond what they would be in the actual deployment context.

The more operationally meaningful comparison is PPV and F1. Option A’s PPV of 0.700 means three in ten flagged patients are false positives, compared to one in ten under Option B; since a sepsis alert model is activated after infection has been suspected, not across all admissions indiscriminately, Option B’s PPV of 0.901 is the figure that would actually govern clinical workload, and its F1 advantage (0.899 vs 0.817) confirms more balanced performance at the chosen threshold. These results confirm that Option B is the appropriate primary analysis dataset and that the primary findings are robust rather than artefacts of the specific cohort definition.

SHAP consistency across cohorts. The core features driving predictions were consistent across both cohorts: lactate_max, platelets_min, ddimer_max, inr_max, and glucose_max all appeared in the top ten for both configurations. The

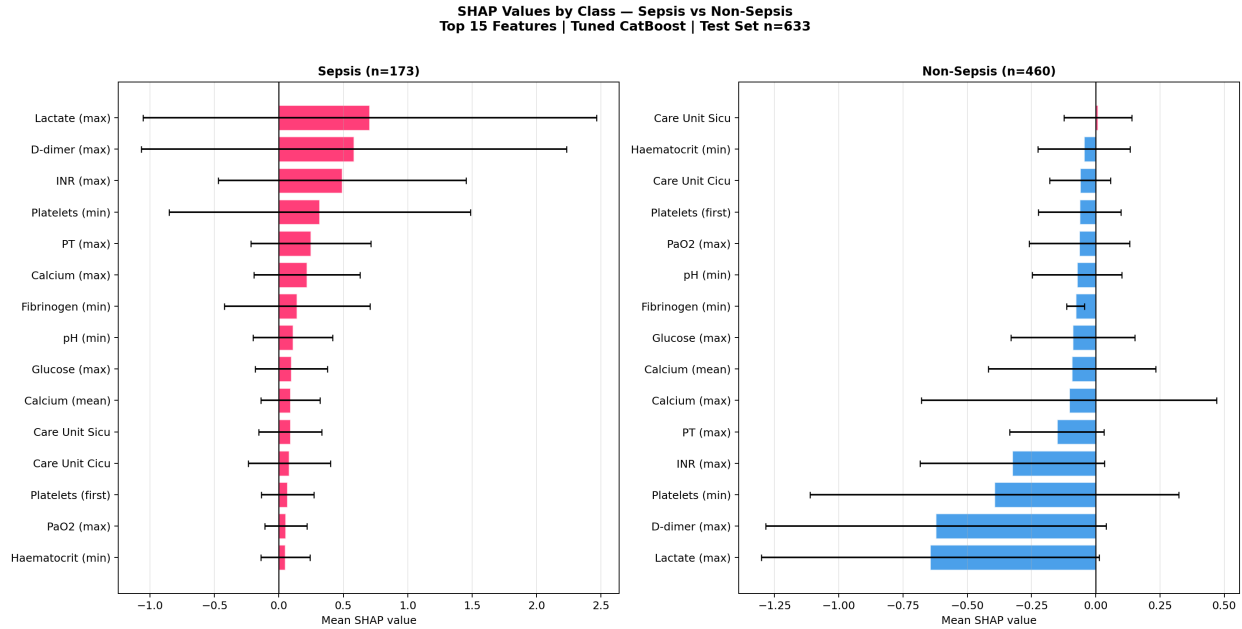


Figure 23: Mean absolute SHAP contributions stratified by true class (sepsis vs non-sepsis), showing which features drive the separation between classes most strongly. The coagulation and cardiovascular features dominate in both directions.

Table 18: Performance comparison between the primary Option B (infection cohort) and sensitivity analysis Option A (full cohort) models at the fixed threshold $t = 0.365$. Delta = Option A minus Option B.

Metric	Option B	Option A	Delta	Interpretation
AUROC	0.9868	0.9968	+0.010	A marginally higher
AUPRC	0.9691	0.9644	-0.005	B marginally higher
Sensitivity	0.896	0.983	+0.087	A: fewer missed sepsis
Specificity	0.963	0.968	+0.005	Near identical
PPV	0.901	0.700	-0.201	A: lower precision
NPV	0.961	0.999	+0.038	A: stronger rule-out
F1	0.899	0.817	-0.082	B: better balanced
Brier Score	0.0385	0.0164	-0.022	A: lower raw calibration error
Training n	2,530	9,957	+7,427	A much larger
Sepsis prevalence	27.3%	6.9%	-20.4%	B more balanced

primary difference was elevated importance of antibiotic-count and microbiology-culture-count features in the Option A model, where the broader non-infected population makes treatment-related variables more discriminative. This reinforces the leakage-aware feature selection decision made for the primary analysis: in the infection-only cohort these treatment proxies carry less additional signal once infection is already established, and including them would risk reintroducing proxy leakage in a subtler form.

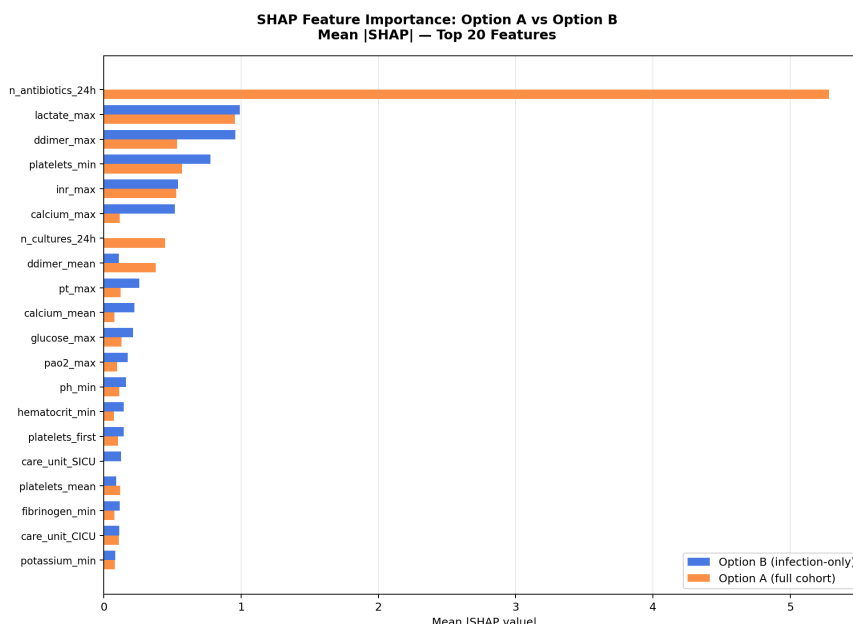


Figure 24: Side-by-side comparison of top-10 SHAP feature importances for the Option A and Option B models. The core physiological features are consistent across both cohorts; treatment-proxy features (antibiotic count, culture count) rise in importance in Option A due to the broader non-infected comparison group.

4. Discussion

This study presents the first published combination of CatBoost, Phoenix Sepsis Criteria labelling, TreeSHAP explainability, and the PIC database in a single pipeline for early paediatric sepsis prediction, alongside the CBFTTS-SA model, a CatBoost-FT-Transformer stacking ensemble developed as this study’s forward-looking architecture. The tuned CatBoost model achieved AUROC 0.9868 and AUPRC 0.9691 on a held-out test set of 633 patients, with strong calibration (Brier Score 0.0385) and clinically actionable operating characteristics at the recommended threshold; the CBFTTS-SA model was statistically indistinguishable from it (AUROC 0.9851, $p = 0.388$). These figures compare favourably with the existing literature: they are consistent with the 0.98 AUROC reported by Chanci et al. using CatBoost and Phoenix labelling on proprietary US hospital data [9], and exceed the 0.916 AUROC reported by Le et al. in a UCSF PICU cohort using a four-hour prediction horizon [4], though differences in labelling standard, prediction window, and cohort composition complicate direct comparison. Across the baseline comparison, CatBoost, LightGBM, and XGBoost all exceeded AUROC 0.97, consistent with broader evidence that gradient boosted trees dominate on moderate-sized clinical tabular data with non-linear feature interactions and class imbalance [12, 10]. The leakage audit is a further methodological contribution: removing vasoactive medication indicators, measurement frequency features, deduplicated measurement- indicator columns, and a length-of-stay outcome proxy eliminated 80 features in total, and the isolation AUROC of the removed count features alone reached 0.878, confirming that models retaining such variables would have systematically overstated predictive capability, a concrete response to the leakage problem Deng et al. identified as pervasive and underreported in clinical machine learning [11].

Benchmarking three deep learning architectures against CatBoost produced a consistent picture. The FT-Transformer achieved AUROC 0.9577, a gap of 0.029 relative to CatBoost that did not reach statistical significance on this test set

($p = 0.169$), in line with Gorishniy et al., who showed that FT-Transformer becomes competitive with gradient boosting only at larger training scale [12]; with 2,530 training patients, the present cohort sits in the moderate-size regime where CatBoost’s ordered-boosting regularisation retains an advantage. The CBFTTS-SA model’s result reinforces this reading: stacking was statistically neutral relative to CatBoost alone, and the meta-learner assigned three to four times more weight to CatBoost than to the FT-Transformer, indicating the transformer component is not yet contributing its full share at this training size. The CBFTTS-SA model is therefore reported as this study’s architecture with the strongest scaling potential rather than its best current performer, while tuned CatBoost remains the stronger and more immediately deployable model today. A more clinically consequential finding is the specificity gap between CatBoost and the deep learning models: all three matched or exceeded CatBoost’s sensitivity but at substantially lower specificity, with TabNet as low as 0.770 against CatBoost’s 0.963 at comparable sensitivity, translating to roughly four times as many false alerts per true positive at this cohort’s 27.3% prevalence. Given that alert fatigue from low-specificity tools is a recognised mechanism through which clinical decision support systems lose clinician trust [21], this balance of sensitivity and specificity, combined with lower calibration error and simpler training requirements, makes CatBoost the operationally preferable choice for this application.

The SHAP analysis indicates that the model’s decision logic is grounded in established pathophysiological markers rather than statistical artefacts. The top five features, lactate, D-dimer, platelets, INR, and PT, map directly onto the cardiovascular and coagulation Phoenix domains that define the sepsis labels in this cohort [3, 2], and the interaction analysis identifies two mechanistically coherent axes: a lactate-platelets interaction reflecting simultaneous circulatory failure and consumptive thrombocytopenia, and an INR-D-dimer coagulopathy axis reflecting concurrent clotting factor depletion and fibrin degradation. The non-monotonic calcium relationship identified in the dependence analysis, in which both hypocalcaemia and hypercalcaemia are associated with increased risk, would not be recoverable by logistic regression and illustrates how gradient boosted trees can surface genuine clinical signal from raw feature values without explicit engineering. The synthetic augmentation results are consistent with this reliance on genuine signal: performance climbed monotonically from synthetic-only (0.929) through hybrid (0.978) to real-data-only (0.986) training, confirming that real data, not synthetic data, drives performance, since the Iman-Conover correction repairs CTGAN’s correlation collapse but cannot manufacture the extreme-tail lactate and D-dimer values that carry the strongest sepsis signal. The age-stratified analysis further supports the model’s robustness: neonates showed the most restrained but still strong performance (AUROC 0.970), consistent with the physiological distinctiveness of the neonatal period, while the infant and child subgroups, comprising 80% of the test cohort, both exceeded AUROC 0.987, indicating that performance is not artificially inflated by concentration in a single age stratum.

5. Conclusion

This study establishes a complete and reproducible machine learning pipeline for early paediatric sepsis prediction on the PIC database, using Phoenix Sepsis Criteria as the labelling standard throughout. A tuned CatBoost model achieves AUROC 0.9868 and AUPRC 0.9691, the strongest reported performance for Phoenix-labelled paediatric sepsis prediction on this database to the authors’ knowledge, at a recommended clinical operating threshold of 0.365 (90% sensitivity, 96.3% specificity). A systematic pre-training leakage audit removed four structurally leaky feature categories measurement-frequency, vasoactive-medication, measurement-indicator, and length-of-stay columns from a 305-feature structurally-cleaned candidate pool, offering a methodological template for future work on the same database. Benchmarking against three deep learning architectures shows the FT-Transformer is statistically non-inferior in discrimination but carries a substantially higher false-positive burden; the CBFTTS-SA model, a CatBoost–FT-Transformer stacking ensemble, performs comparably to CatBoost alone ($p = 0.388$) and is presented as the more promising architecture as paediatric ICU datasets grow, while CatBoost remains the operationally preferable choice today. A CTGAN augmentation pipeline with Iman-Conover correlation correction shows that real data, not synthetic data, drives performance in this setting. TreeSHAP analysis further grounds the model’s decision logic in established paediatric sepsis pathophysiology, supporting clinician trust in its outputs.

These contributions address three concrete gaps in the existing literature: the absence of Phoenix-labelled paediatric sepsis prediction on reproducible open data, the absence of rigorous leakage controls in published PIC-based models, and the absence of an interpretable layer clinicians can engage with in deployment contexts. The findings should be read alongside the constraints of a retrospective, single-centre design with a modest events-per-variable ratio and a labelling scheme limited to two of the four Phoenix domains by the data available. Prospective, multi-site

external validation is accordingly the necessary next step before any clinical deployment, ideally paired with a time-series reformulation that evaluates risk continuously rather than at a single fixed window and, as larger paediatric ICU datasets become available, re-evaluation of the CBFTTS-SA model's scaling advantage. The methodological foundation established here, from leakage-audited features through calibrated, interpretable predictions, provides a platform for that work.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability

The PIC database is publicly available via PhysioNet at <https://physionet.org>. All preprocessing and modelling code is available at <https://github.com/tanishqkr/pediatric-sepsis-PIC>.

Ethics Statement

This study used the Paediatric Intensive Care (PIC) database (version 1.1.0), a de-identified, publicly available dataset accessible via PhysioNet under credentialed researcher access. No identifiable patient data were used or accessed. In accordance with the institutional guidelines of SVKM's NMIMS Indore Campus and the terms of the PhysioNet Credentialed Health Data Use Agreement, no additional Institutional Review Board approval was required for this retrospective analysis of pre-existing, de-identified data.

References

- [1] A. Kumar, D. Roberts, K. E. Wood, B. Light, J. E. Parrillo, S. Sharma, et al., Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock, *Critical Care Medicine* 34 (6) (2006) 1589–1596. doi:10.1097/01.CCM.0000217961.75225.E9.
- [2] L. J. Schlapbach, R. S. Watson, L. R. Sorce, A. C. Argent, K. Menon, M. W. Hall, et al., International consensus criteria for pediatric sepsis and septic shock, *JAMA* 331 (8) (2024) 665–674. doi:10.1001/jama.2024.0179.
- [3] L. N. Sanchez-Pinto, T. D. Bennett, P. E. DeWitt, S. Russell, M. N. Rebull, B. Martin, et al., Development and validation of the Phoenix criteria for pediatric sepsis and septic shock, *JAMA* 331 (8) (2024) 675–686. doi:10.1001/jama.2024.0196.
- [4] S. Le, J. Hoffman, C. Barton, J. C. Fitzgerald, A. Allen, E. Pellegrini, et al., Pediatric severe sepsis prediction using machine learning, *Frontiers in Pediatrics* 7 (2019) 413. doi:10.3389/fped.2019.00413.
- [5] Z. Yang, X. Cui, Z. Song, Predicting sepsis onset in ICU using machine learning models: A systematic review and meta-analysis, *BMC Infectious Diseases* 23 (1) (2023) 635. doi:10.1186/s12879-023-08614-0.
- [6] B. Goldstein, B. Giroir, A. Randolph, International pediatric sepsis consensus conference: Definitions for sepsis and organ dysfunction in pediatrics, *Pediatric Critical Care Medicine* 6 (1) (2005) 2–8. doi:10.1097/01.PCC.0000149131.72248.E6.
- [7] S. Leteurtre, A. Duhamel, J. Salleron, B. Grandbastien, J. Lacroix, F. Leclerc, PELOD-2: An update of the PEdiatric logistic organ dysfunction score, *Critical Care Medicine* 41 (7) (2013) 1761–1773. doi:10.1097/CCM.0b013e31828a2bbd.

- [8] X. Zeng, G. Yu, Y. Lu, L. Tan, X. Wu, S. Shi, et al., A study of the Paediatric Intensive Care (PIC) database, *Scientific Data* 7 (2020) 184. doi:10.1038/s41597-020-0355-4.
- [9] D. Chanci, J. R. Grunwell, A. Rafiei, S. R. Brown, M. J. Ripple, N. R. Bishop, et al., Machine learning model for daily prediction of pediatric sepsis using Phoenix criteria, *Pediatric Research* (2025). doi:10.1038/s41390-025-04221-8.
- [10] L. M. Fleuren, T. L. T. Klausch, C. L. Zwager, L. J. Schoonmade, T. Guo, L. F. Roggeveen, et al., Machine learning for the prediction of sepsis: A systematic review and meta-analysis of diagnostic test accuracy, *Intensive Care Medicine* 46 (3) (2020) 383–400. doi:10.1007/s00134-019-05872-y.
- [11] H.-F. Deng, M.-W. Sun, Y. Wang, J. Zeng, T. Yuan, T. Li, et al., Evaluating machine learning models for sepsis prediction: A systematic review of methodologies, *iScience* 25 (2) (2022) 103651. doi:10.1016/j.isci.2021.103651.
- [12] Y. Gorishniy, I. Rubachev, V. Khrulkov, A. Babenko, Revisiting deep learning models for tabular data, in: *Advances in Neural Information Processing Systems*, Vol. 34, 2021, pp. 18932–18943. URL <https://proceedings.neurips.cc/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-Abstract.html>
- [13] H.-J. Han, K. Kim, J. D. Park, Early detection of bloodstream infection in critically ill children using artificial intelligence, *Acute and Critical Care* 39 (4) (2024) 611–620. doi:10.4266/acc.2024.00752.
- [14] X. Shi, X. Wang, H. Yang, X. Fan, G. Liu, Y. Song, et al., Accurate prediction of sepsis from pediatric emergency department to PICU using a machine-learning model, *Frontiers in Pediatrics* 13 (2025) 1610187. doi:10.3389/fped.2025.1610187.
- [15] S. M. Lauritsen, M. Kristensen, M. V. Olsen, M. S. Larsen, K. M. Lauritsen, M. J. Jørgensen, et al., Explainable artificial intelligence model to predict acute critical illness from Electronic Health Records, *Nature Communications* 11 (2020) 3852. doi:10.1038/s41467-020-17431-x.
- [16] M. Moor, M. Horn, B. Rieck, D. Roqueiro, K. Borgwardt, Early recognition of sepsis with gaussian process temporal convolutional networks and dynamic time warping, in: *Proceedings of the 4th Machine Learning for Healthcare Conference*, Vol. 106 of *Proceedings of Machine Learning Research*, PMLR, 2019, pp. 2–26. URL <https://proceedings.mlr.press/v106/moor19a.html>
- [17] R. Kamaleswaran, O. Akbilgic, M. A. Hallman, A. N. West, R. L. Davis, S. H. Shah, Applying artificial intelligence to identify physiometers predicting severe sepsis in the pediatric intensive care unit, *Pediatric Critical Care Medicine* 19 (10) (2018) e495–e503. doi:10.1097/PCC.0000000000001666.
- [18] S. O. Arik, T. Pfister, TabNet: Attentive interpretable tabular learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35, 2021, pp. 6679–6687. doi:10.1609/aaai.v35i8.16826.
- [19] L. Xu, M. Skoularidou, A. Cuesta-Infante, K. Veeramachaneni, Modeling tabular data using conditional GAN, in: *Advances in Neural Information Processing Systems*, Vol. 32, 2019, pp. 7335–7345. URL <https://proceedings.neurips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html>
- [20] R. L. Iman, W. J. Conover, A distribution-free approach to inducing rank correlation among input variables, *Communications in Statistics — Simulation and Computation* 11 (3) (1982) 311–334. doi:10.1080/03610918208812265.
- [21] V. S. Athukorala, W. M. K. S. Ilmini, Explainable AI for critical care: A systematic review of interpretable models for sepsis and ICU mortality prediction, *BMC Medical Informatics and Decision Making* 26 (2026) 64. doi:10.1186/s12911-026-03344-0.

- [22] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems*, Vol. 30, 2017, pp. 4765–4774.
URL <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [23] F. Shen, Y. Xu, X. Jiang, L. Yu, H.-H. Ge, X. Wang, et al., Predicting outcomes in pediatric patients with acute kidney injury: A retrospective single-center cohort study using machine learning models, *BMC Medical Informatics and Decision Making* 25 (2025) 371. doi:10.1186/s12911-025-03211-4.
- [24] J. Zhou, Y. Xie, Y. Liu, P. Niu, H. Chen, X. Zeng, et al., Interpretable machine learning model for early prediction of disseminated intravascular coagulation in critically ill children, *Scientific Reports* 15 (2025) 11217. doi:10.1038/s41598-025-91434-w.
- [25] Q. Huang, L. Zheng, R. Cai, H. Chen, Machine learning-based time-to-event survival analysis in pediatric patients with severe sepsis, *Frontiers in Pediatrics* 13 (2025) 1688416. doi:10.3389/fped.2025.1688416.
- [26] J. Yang, H. Peng, Y. Luo, T. Zhu, L. Xie, Explainable ensemble machine learning model for prediction of 28-day mortality risk in patients with sepsis-associated acute kidney injury, *Frontiers in Medicine* 10 (2023) 1165129. doi:10.3389/fmed.2023.1165129.
- [27] S. Hong, X. Hou, J. Jing, W. Ge, L. Zhang, Predicting risk of mortality in pediatric ICU based on ensemble step-wise feature selection, *Health Data Science 2021* (2021) 9365125. doi:10.34133/2021/9365125.
- [28] Y. Ding, T. Mei, Triglyceride-glucose index and 30-day mortality in pediatric sepsis: A retrospective cohort study based on the PIC database, *Frontiers in Pediatrics* 14 (2026) 1704208. doi:10.3389/fped.2026.1704208.
- [29] G. S. Collins, J. B. Reitsma, D. G. Altman, K. G. M. Moons, Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement, *Annals of Internal Medicine* 162 (1) (2015) 55–63. doi:10.7326/M14-0697.